Towards Generalizable Safety for Robot Autonomy

Combining Model-based Analysis with Data-driven Approaches

Jason J. Choi · Ph.D. Seminar, UC Berkeley, April 16, 2025
Video: https://www.youtube.com/watch?v=GYWpmFHwGFg

THE TALK IN 5 POINTS

- **Problem:** safety guarantees from control theory need an accurate dynamics model *f*. New vehicles such as eVTOL air taxis don't have one.
- **Indirect way** (most of the PhD): start from a rough model, learn its error from data (GP, RL), then build robust safe sets or CBF filters. It works, but needs a lot of engineering and can be very conservative.
- **Direct way** (new): the **Data-Driven Hamiltonian (DDH)** runs HJ reachability straight from (state, control, velocity) data. It works with sets of possible velocities and uses the worst case inside an uncertainty ball around each data point.
- **Guarantee:** $\hat{H} \leq H$, so the data-driven safe set sits inside the true safe set. It never calls an unsafe state safe.
- **XV-15 tiltrotor demo:** safe data collection over 19 rounds recovers 51.6% of the true safe set volume.

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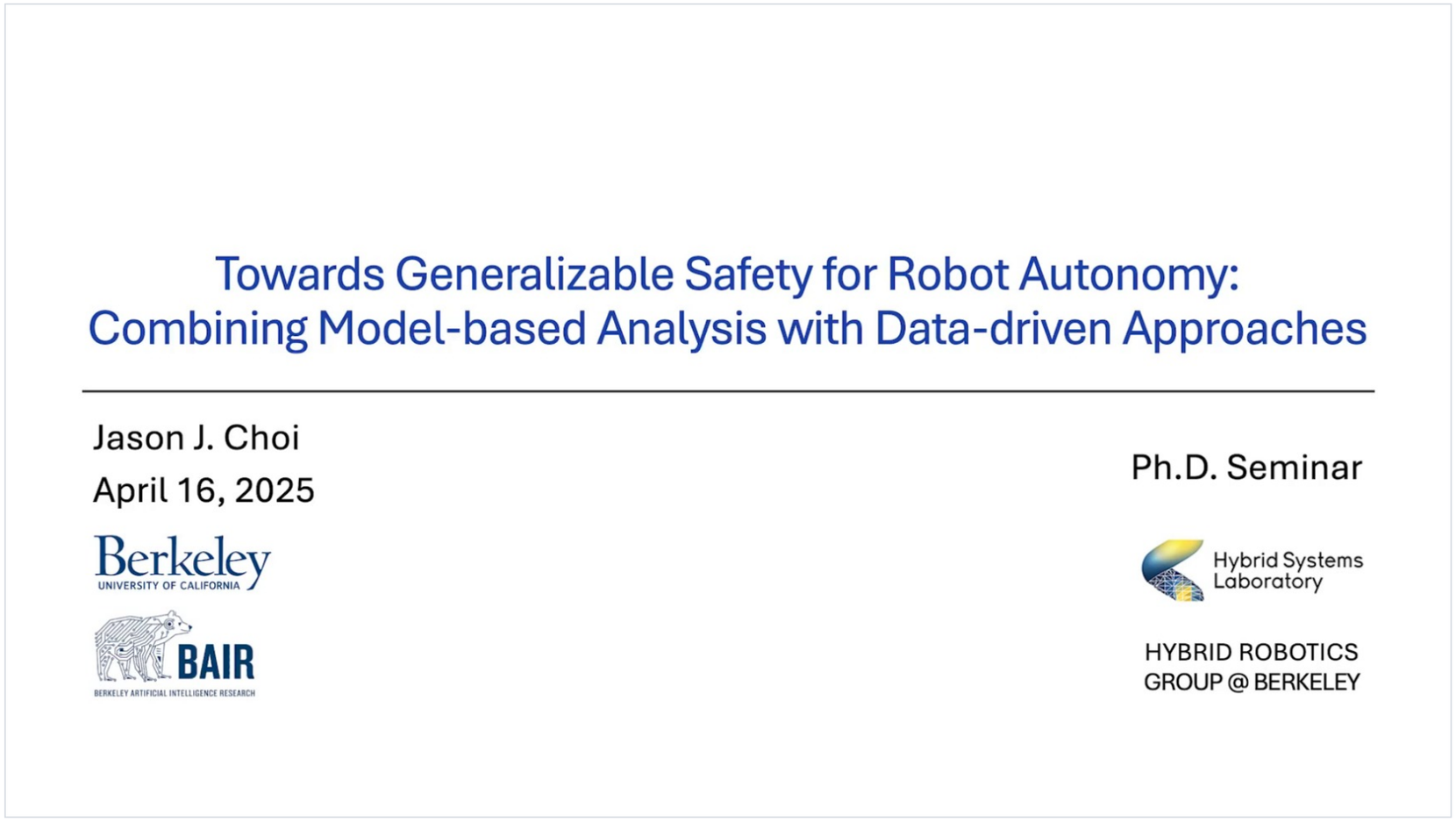
Safe set: states from which the robot can always avoid failure.

HJ reachability: computes the largest safe set by solving a PDE (a partial differential equation) for a value function V; safe where $V \geq 0$.

Hamiltonian H(x,p): the term in the HJ PDE where the dynamics enter: max over u of $p^T f(x,u)$.

CBF (control barrier function): a function B whose rate of change is kept bounded so the robot stays where $B \geq 0$.

Safety filter: changes a controller's command only when needed to stay safe.

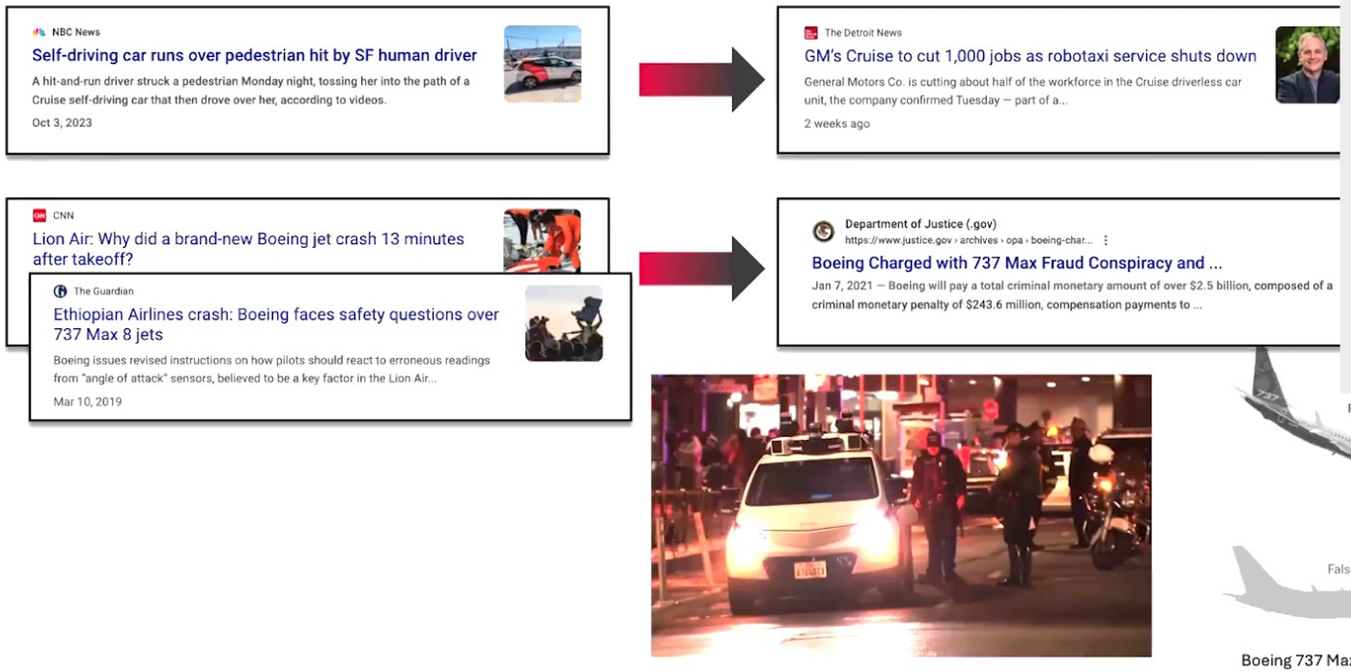


1. BACKGROUND & MOTIVATION

Title and introduction

▶ 6:51 · intro from 0:00

- Jason J. Choi's PhD seminar at UC Berkeley, co-advised by Koushil Sreenath and Claire Tomlin.
- The first ~7 min (room camera, no slides) are the advisors' introduction. His first big project linked HJ reachability and control barrier functions. Later work: data-driven safety filters with NASA and Joby, and safe multi-agent RL.
- Talk goal: safety methods that work across many kinds of robots, by combining model-based control theory with data-driven methods.

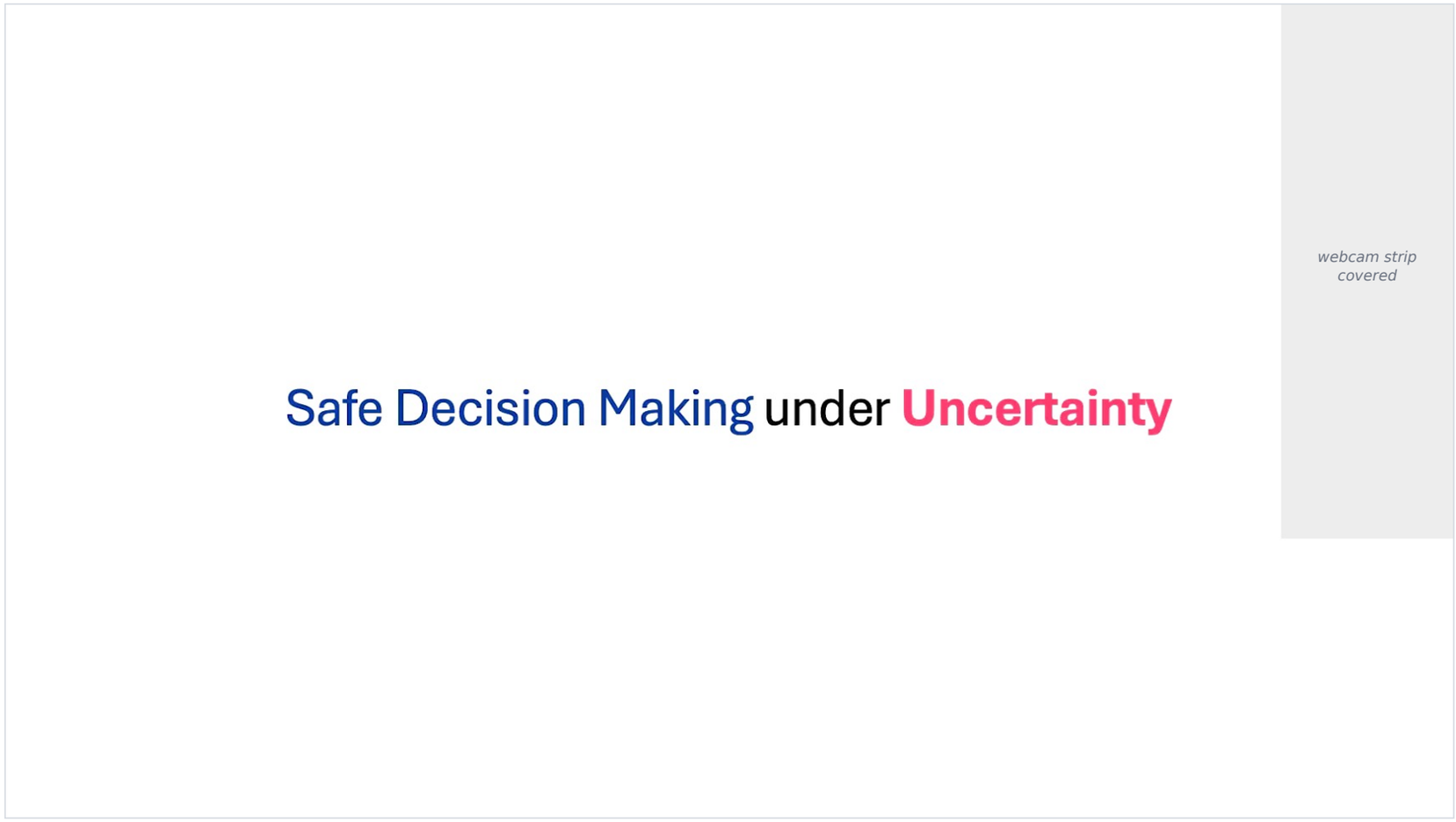


1. BACKGROUND & MOTIVATION

Why safety guarantees matter

▶ 7:37

- Long-term goal: autonomous machines that help society. A key hurdle is proving that they are safe.
- At large scale, even rare failures are costly. Example 1: a Cruise robotaxi ran over a pedestrian (who had first been hit by a human driver), and Cruise later shut down its robotaxi service.
- Example 2: the Lion Air and Ethiopian Airlines 737 MAX crashes, linked to a false angle-of-attack sensor reading (the plane animation), after which Boeing was charged with fraud conspiracy.
- Guaranteed safety is the foundation for reliable long-term operation. Building such guarantees is the focus of the PhD.

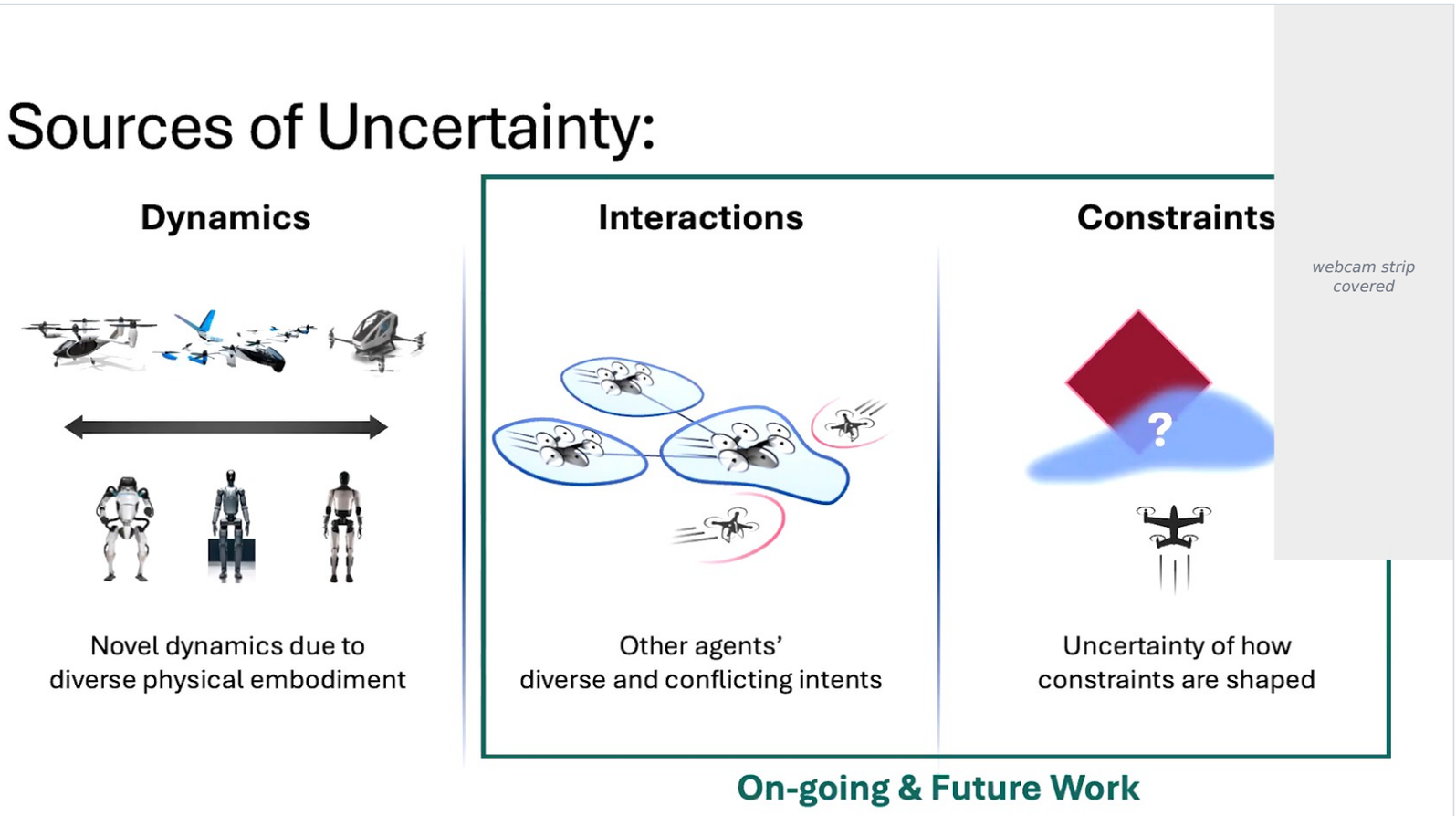


1. BACKGROUND & MOTIVATION

Safe decision making under uncertainty

▶ 8:57

- Why it's hard: real-world decisions face many kinds of uncertainty, and any of them can break a safety guarantee.



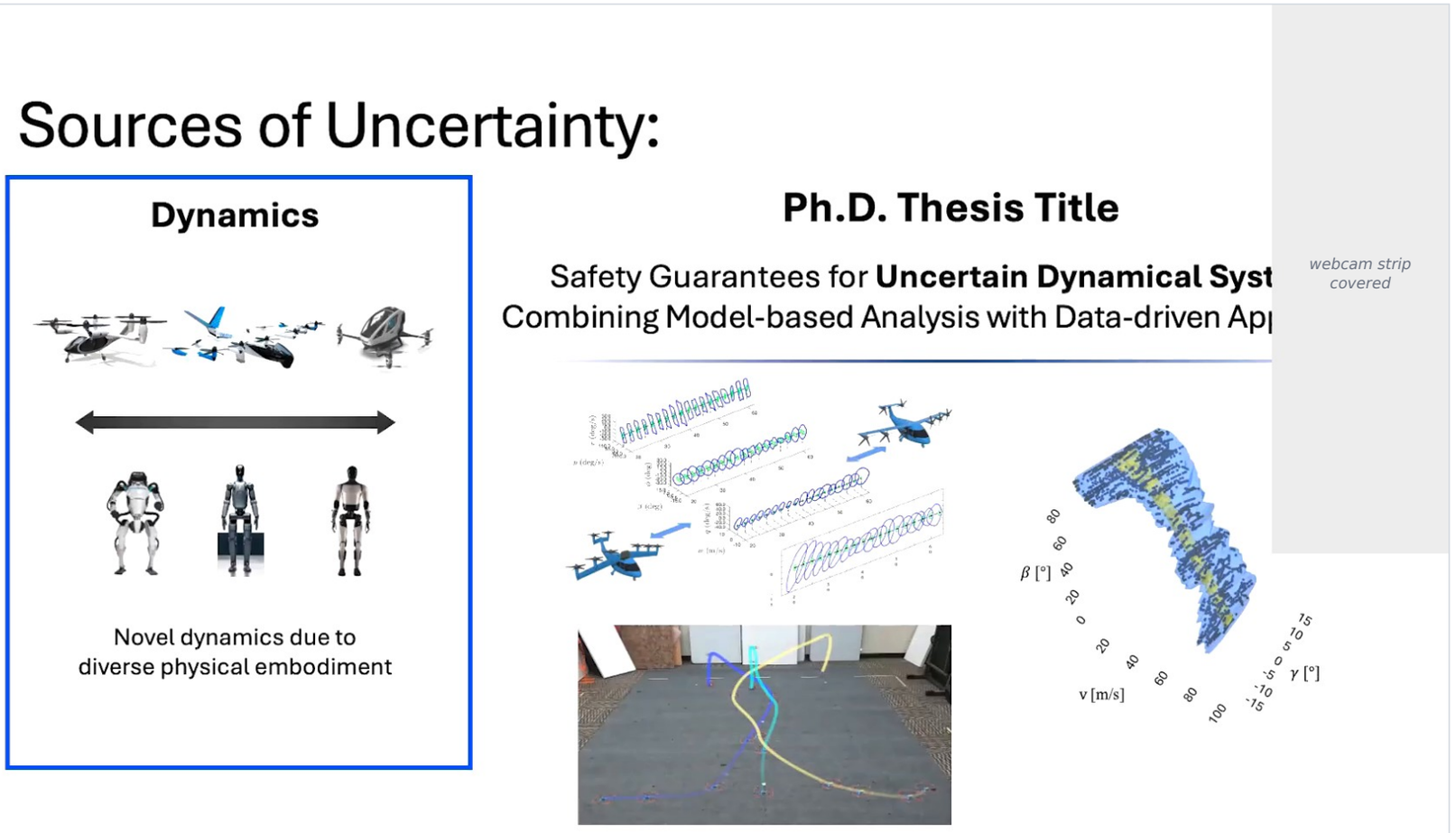
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1. BACKGROUND & MOTIVATION

Three sources of uncertainty

▶ 9:08

- **Dynamics:** each new body (air taxis, humanoids) moves differently. If the model is wrong, the guarantee can fail.
- **Interactions:** other agents have unknown or conflicting goals.
- **Constraints:** perception errors, partial observability, and limits that are hard to write as math.
- Interactions and constraints are ongoing and future work. The PhD focuses on uncertain dynamics.



1. BACKGROUND & MOTIVATION

Thesis focus: uncertain dynamics

▶ 11:06

- Thesis: *Safety Guarantees for Uncertain Dynamical Systems: Combining Model-based Analysis with Data-driven Approaches.*
- Treat any robot as a dynamical system whose equations are not fully known. Combine control-theory tools with data to certify safety.

Outline

1. Background & Motivation

2. **Indirect Approaches for Data-Driven Safety**

3. **Direct Approach: Data-Driven Hamiltonian (DDH)**

4. Other works & Future research directions

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1. BACKGROUND & MOTIVATION

Outline

▶ 11:42

- (1) Background and motivation.
- (2) Indirect data-driven safety: a quick tour of past work.
- (3) Direct approach, the Data-Driven Hamiltonian (DDH): new work submitted to CDC 2025, covered in depth.
- (4) Other work and future directions.

Outline

1. **Background & Motivation**
   - **Fundamentals for safe decision making**
   - **Hamilton-Jacobi reachability framework**
   - **Novel dynamics for new generation aerial vehicles**
2. Indirect Approaches for Data-Driven Safety
3. Direct Approach: Data-Driven Hamiltonian (DDH)
4. Other works & Future research directions

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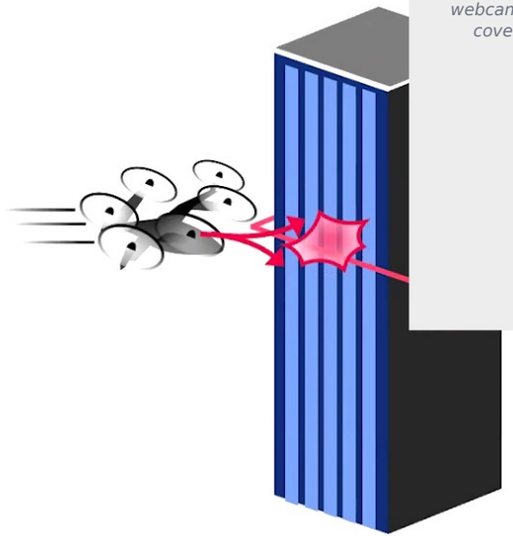
1. BACKGROUND & MOTIVATION

Part 1: Background & motivation

▶ 12:19

- Covers the basics of safe decision making, the Hamilton-Jacobi (HJ) reachability framework, and why new aerial vehicles need data-driven safety.

1.1 Fundamental Problems for Safety



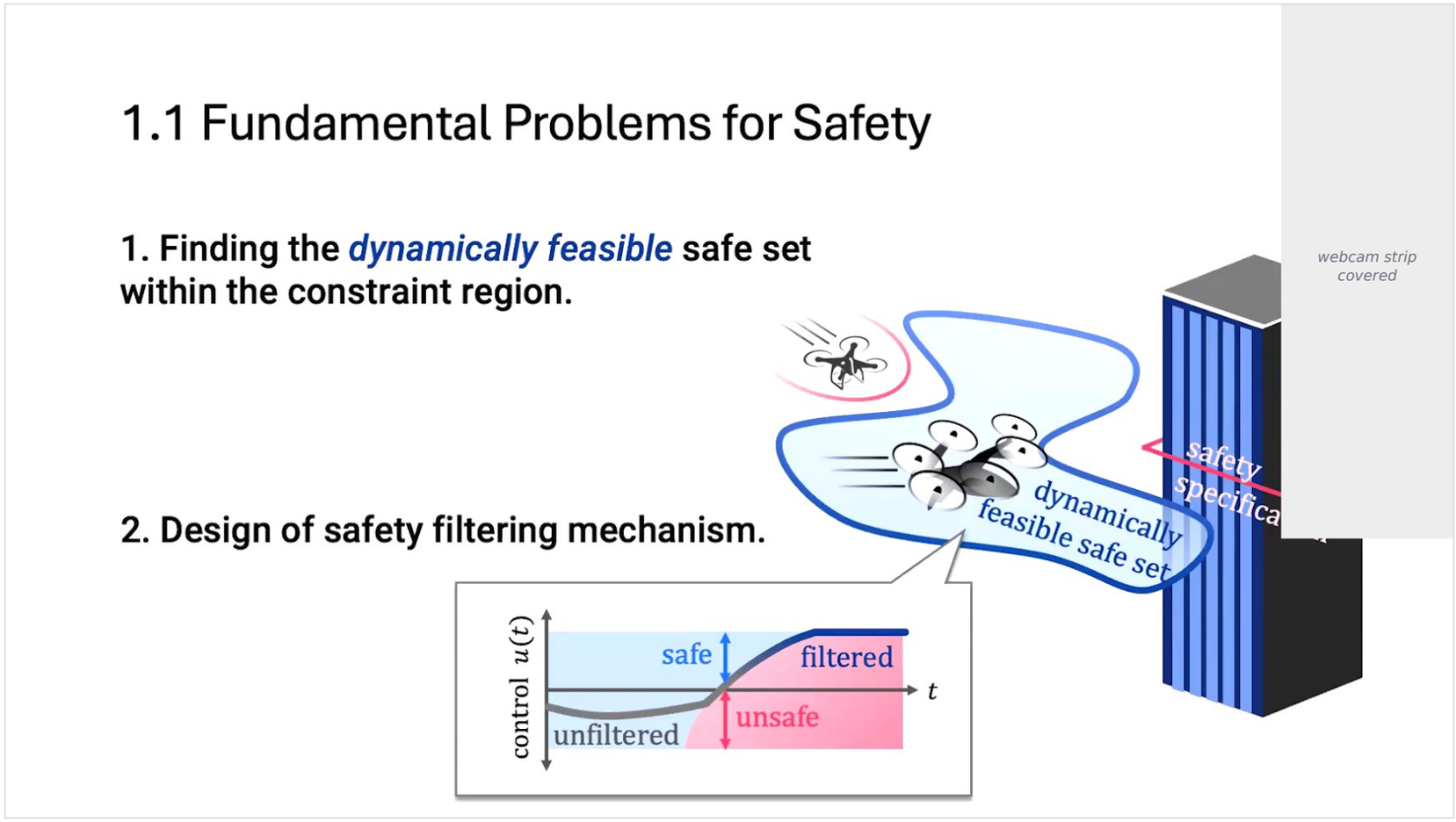
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1. BACKGROUND & MOTIVATION

Not every allowed state is safe

▶ 12:24

- Example: an air taxi must not hit a building. Avoiding the building is the safety constraint.
- Some states inside the allowed region are already doomed: no control input can stop the crash.
- So the allowed region is bigger than the truly safe region.

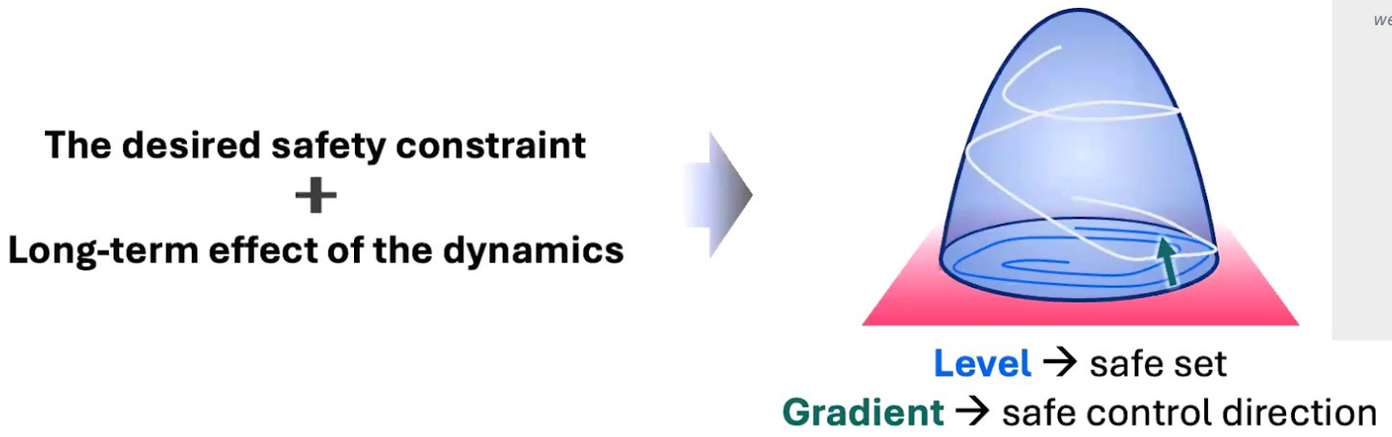


1. BACKGROUND & MOTIVATION

Two fundamental problems

▶ 13:06

- **1.** Find the *dynamically feasible safe set*: the part of the allowed region the robot can always stay in.
- **2.** Design a **safety filter**: at each state, pick inputs that keep the robot in that set. The plot shows the filter changing the input only when it would leave the safe band.
- Solving both gives a controller with a safety guarantee.



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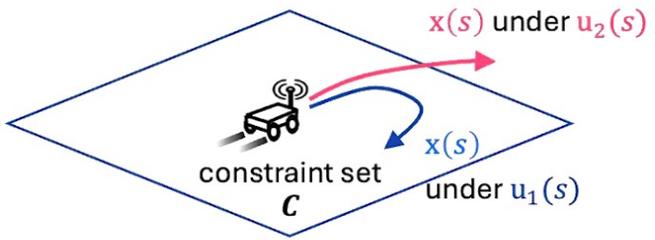
1. BACKGROUND & MOTIVATION

Certificate functions for safety

▶ 13:48

- A certificate function is one scalar function that combines the safety constraint with the long-term effect of the dynamics.
- Its **level** marks the safe set (e.g. safe where $V(x) \geq 0$). Its **gradient** tells which control directions are safe.
- If you can compute one, you can both verify safety and steer the robot to stay safe.

1.2 Hamilton-Jacobi (HJ) Reachability Framework



$$\dot{\mathrm{x}}(s) = f(\mathrm{x}(s), \mathrm{u}(s))$$

dynamics — control signal

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1. BACKGROUND & MOTIVATION

HJ reachability: setup

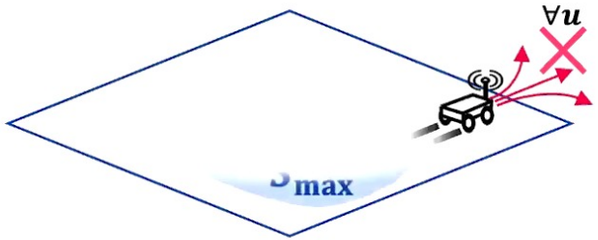
▶ 14:27

- HJ reachability is a well-established tool (~20 years) for computing certificate functions.
- Setup: the robot must stay inside the box (constraint set *C*), and it moves by $\dot{x} = f(x, u)$.
- Depending on the input (u_1 or u_2), the robot stays inside or leaves the box.

1.2 Hamilton-Jacobi (HJ) Reachability Framework

Main Idea:

Compute **maximal safe set** $S_{\max}$ as an optimal control problem.



Bansal*, Chen*, Herbert*, Tomlin, CDC'17

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1. BACKGROUND & MOTIVATION

HJ reachability: main idea

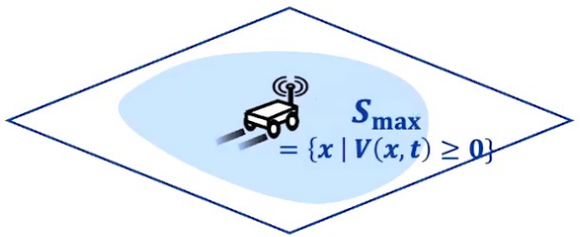
▶ 15:11

- Find the **largest safe set** S_{max} by posing it as an optimal control problem.
- Remove every state from which *all* inputs (∀u) lead out of the box. What is left is S_{max}: states where at least one input keeps the robot inside.

1.2 Hamilton-Jacobi (HJ) Reachability Framework

Main Idea:

Compute **maximal safe set $S_{\max}$** as an optimal control problem.



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Safety Certificate (Value Function):

$$V(x,t) = \max_{u \in \mathcal{U}_{[t;0]}} \min_{s \in [t,0]} g(x(s))$$

opt. ctrl. — closest distance to failure region

Bansal*, Chen*, Herbert*, Tomlin, CDC'17

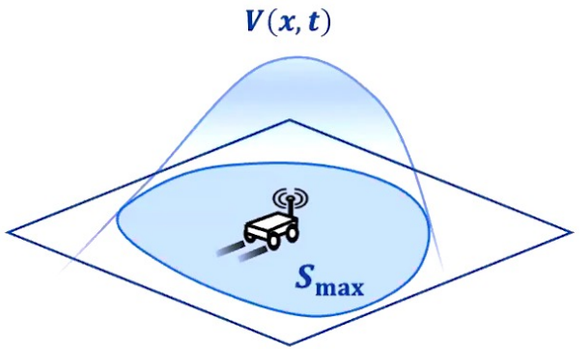
1. BACKGROUND & MOTIVATION

Value function = safety certificate

▶ 15:41

- $g(x)$ = distance to the failure boundary (positive inside). $V(x,t)$ = max over controls of the smallest g along the future path.
- Example: if, under the best control, the path never gets closer than 0.3 m to the wall, then $V = 0.3$ and the state is safe.
- So $S_{max} = \{x : V(x,t) \geq 0\}$, and keeping $V \geq 0$ keeps the robot safe.

1.2 Hamilton-Jacobi (HJ) Reachability Framework



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Solution to Hamilton-Jacobi PDE:

$$0 = \min\{g(x) - V(x,t),\ -D_t V(x,t) + H(x, D_x V(x,t))\}$$

$$V(x,0) = g(x) \quad \text{(Terminal Condition)}$$

Bansal*, Chen*, Herbert*, Tomlin, CDC'17

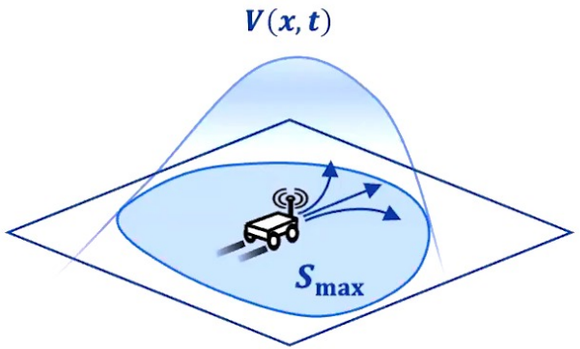
1. BACKGROUND & MOTIVATION

The Hamilton-Jacobi PDE

▶ 16:41

- By dynamic programming, *V* solves the HJ PDE: 0 = min{ g(x) − V, −∂V/∂t + H(x, ∇V) }, with V(x,0) = g(x).
- Finding S_{max} comes down to solving this PDE, usually on a grid.
- The dynamics are hidden inside *H*, the Hamiltonian.

1.2 Hamilton-Jacobi (HJ) Reachability Framework



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Hamiltonian:

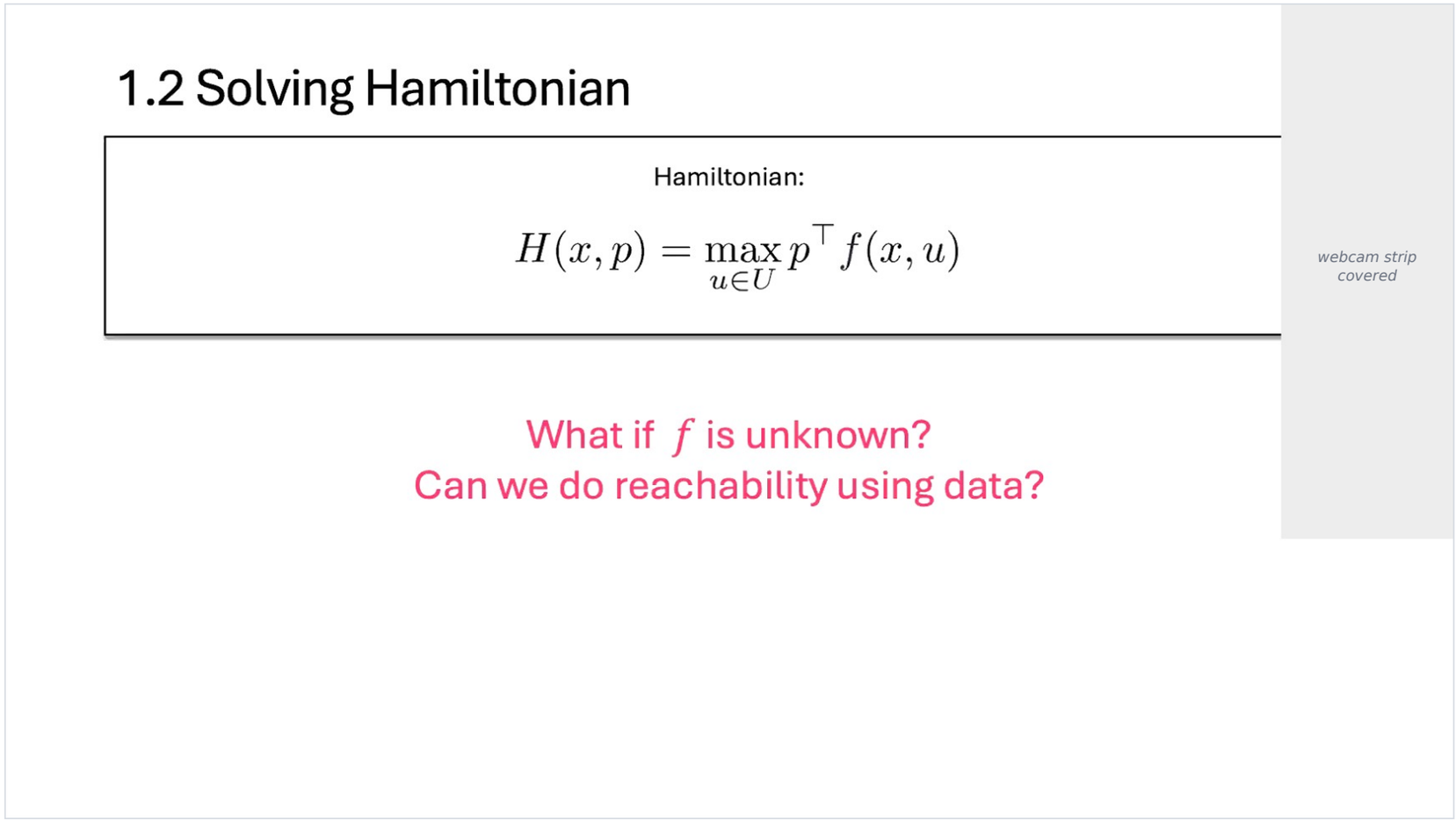
$$H(x,p) = \max_{u \in U} p^\top \underbrace{f(x,u)}_{\text{dynamics}}$$

1. BACKGROUND & MOTIVATION

The Hamiltonian needs the dynamics

▶ 17:22

- H(x,p) = max over u ∈ U of $p^T f(x,u)$.
- *p* is the gradient of *V* (formally the costate). The speaker calls it the "objective direction".
- Evaluating *H* at every grid point requires knowing *f* exactly.



1. BACKGROUND & MOTIVATION

What if f is unknown?

▶ 17:57

- Motivating question: can we do reachability, and safety analysis in general, directly from data without knowing *f*?

1.3 New generation electric vertical takeoff-and-landing vehicles (eVTOLs)

NASA Tiltwing for Urban Air Mobility Concept of Operations



Joby Aviation S4



webcam strip covered

1. BACKGROUND & MOTIVATION

New eVTOL air taxis

▶ 18:07

- Electric vertical take-off and landing (eVTOL) vehicles, e.g. NASA Tiltwing and Joby S4, switch between hover mode and cruise mode by tilting wings or rotors.
- Their dynamics are hard to pin down. Risk peaks during the hover-to-cruise transition, when loss of control is most likely.
- He works with NASA and Joby on this problem.



1. BACKGROUND & MOTIVATION

Loss of control: a real risk

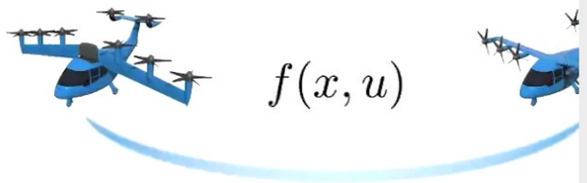
▶ 18:36

- Such accidents have happened in real operations and in flight tests. Shown: a 2017 Osprey crash caused by heavy downwash, and the 2023 Vertical Aerospace VX4 crash.

1.3 Novel & Uncertain Dynamics (f) of eVTOL vehicles

Hamiltonian:

$$H(x,p) = \max_{u \in U} p^\top f(x,u)$$



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- Changing dynamics during their flight mode transition
- Complex aerodynamics hard to model or simulate.
- Lack of closed-form mathematical models
- Lack of special structures (e.g. control-affineness) for easy Hamiltonian
- Extensive data from flight tests

1. BACKGROUND & MOTIVATION

Why eVTOL dynamics are hard

▶ 18:56

- The dynamics change during the mode transition, and rotor-rotor and rotor-airframe aerodynamics are hard to model or simulate.
- So there is no closed-form model, and no helpful structure (e.g. control-affine: $\dot{x} = f(x) + g(x)u$) that makes H easy to solve.
- But flight tests produce lots of trajectory data. Can we use it for safety?

Outline

1. Background & Motivation
2. **Indirect Approaches for Data-Driven Safety**
   - **Data-driven treatment of model error + HJ Reachability / CBF**
3. Direct Approach: Data-Driven Hamiltonian (DDH)
4. Other works & Future research directions

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2. INDIRECT APPROACHES FOR DATA-DRIVEN SAFETY

Part 2: Indirect approaches

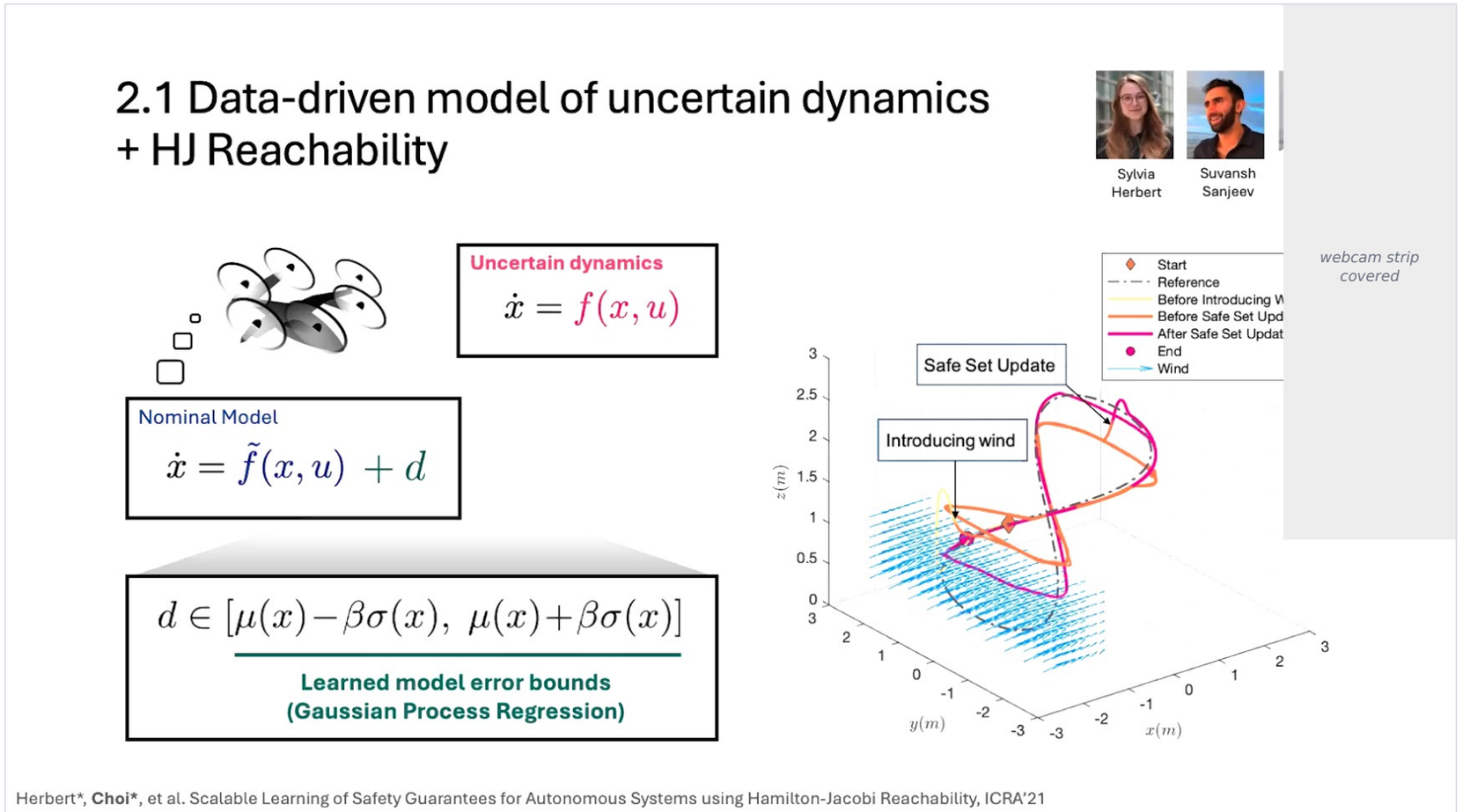
▶ 20:00

- About 80–90% of the PhD falls under this "indirect" approach: learn a separate model of the error, then design for safety.
- Quick tour: 7 papers in 4 slides.

2.1 Learned model error + HJ reachability

▶ 20:31

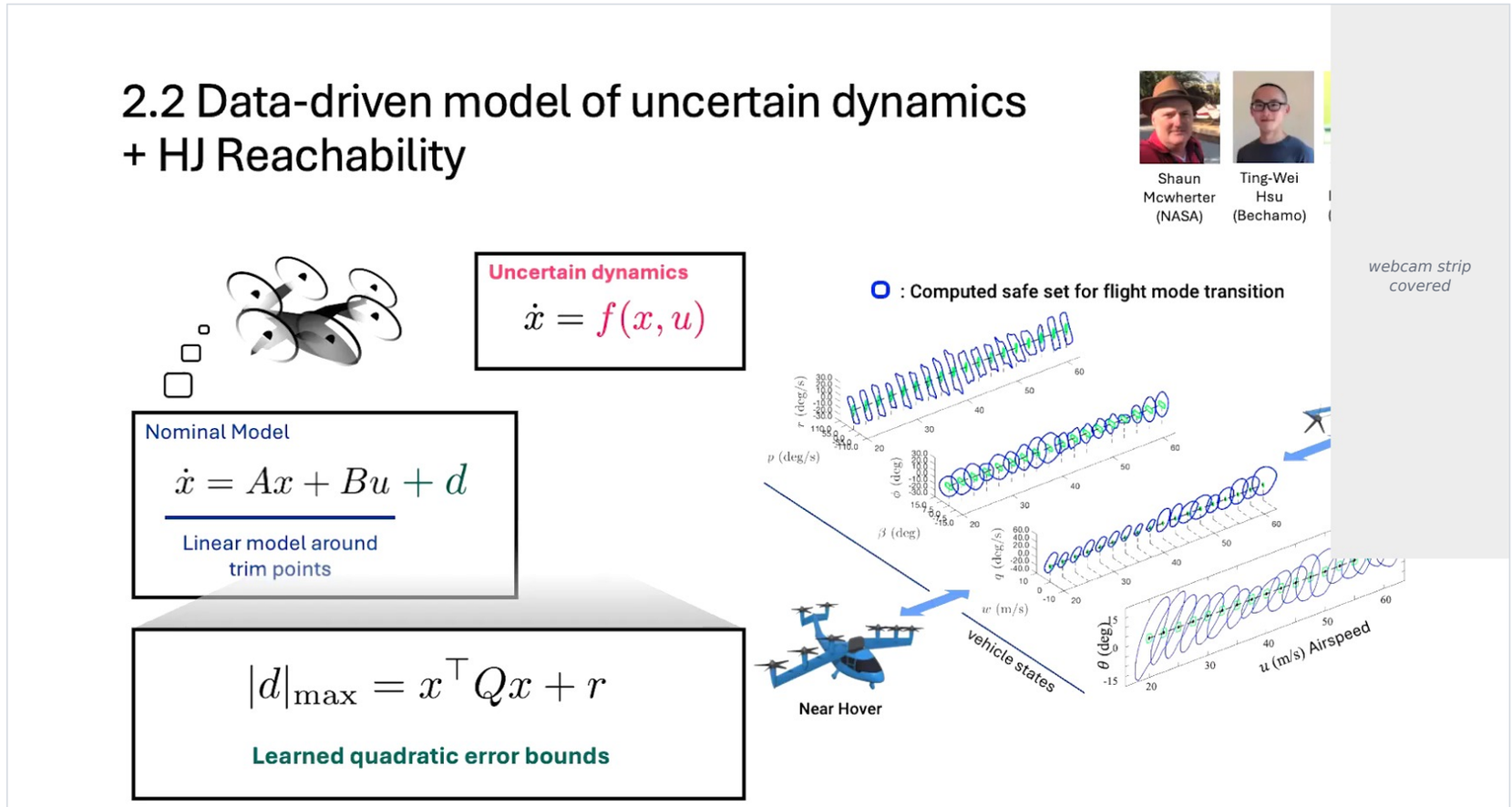
- Start from a nominal model f̃ and add an error term *d* for the gap to the true *f*.
- Learn bounds on *d* from data with Gaussian process (GP) regression: d ∈ [μ − βσ, μ + βσ].
- Compute a safe set that is robust to this error, online. Demo: a quadrotor flying a figure-8 hits wind, does a safety maneuver, learns the disturbance, updates its safe set and continues. (ICRA'21)



2.2 NASA Tiltwing flight envelope

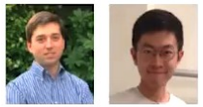
▶ 22:09

- With NASA and Bechamo: verify the safe flight envelope of the NASA Tiltwing through the transition.
- Only Simulink models were available, so they used linear models around trim points: $\dot{x} = Ax + Bu + d$.
- The error bound $|d| \leq x^T Qx + r$ grows fast away from trim, so the verified safe tube is safe but quite conservative. (JAHS'24)



Hsu, **Choi**, et al., Towards Flight Envelope Protection for the NASA Tiltwing eVTOL Flight Mode Transition Using Hamilton-Jacobi Reachability, JAHS'24

2.3 Data-driven model of uncertain dynamics + Control Barrier Function (CBF)


Fernando Castaneda

Bike Zhang



Uncertain dynamics

$$\dot{x} = f(x,u)$$

Nominal Model

$$\dot{x} = \tilde{f}(x,u)$$



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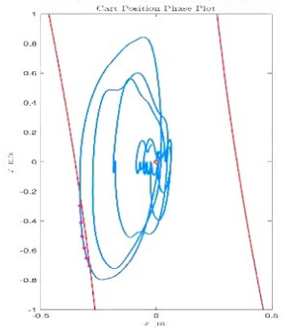
$\dot{B}(x,u) = \nabla B(x) \cdot f(x,u)$:ground-truth (unknown)

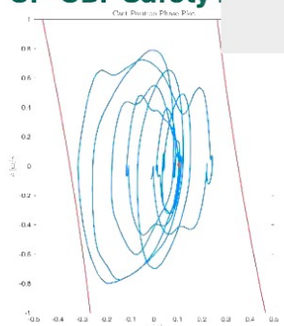
$\tilde{\dot{B}}(x,u) = \nabla B(x) \cdot \tilde{f}(x,u)$:model-based estimate

$$\dot{B}(x,u) = \tilde{\dot{B}}(x,u) + \boldsymbol{\Delta}(x,u)$$

$\widehat{\boldsymbol{\Delta}}(x,u)$: data-driven model of the uncertainty

Nominal model **GP-CBF Safety**





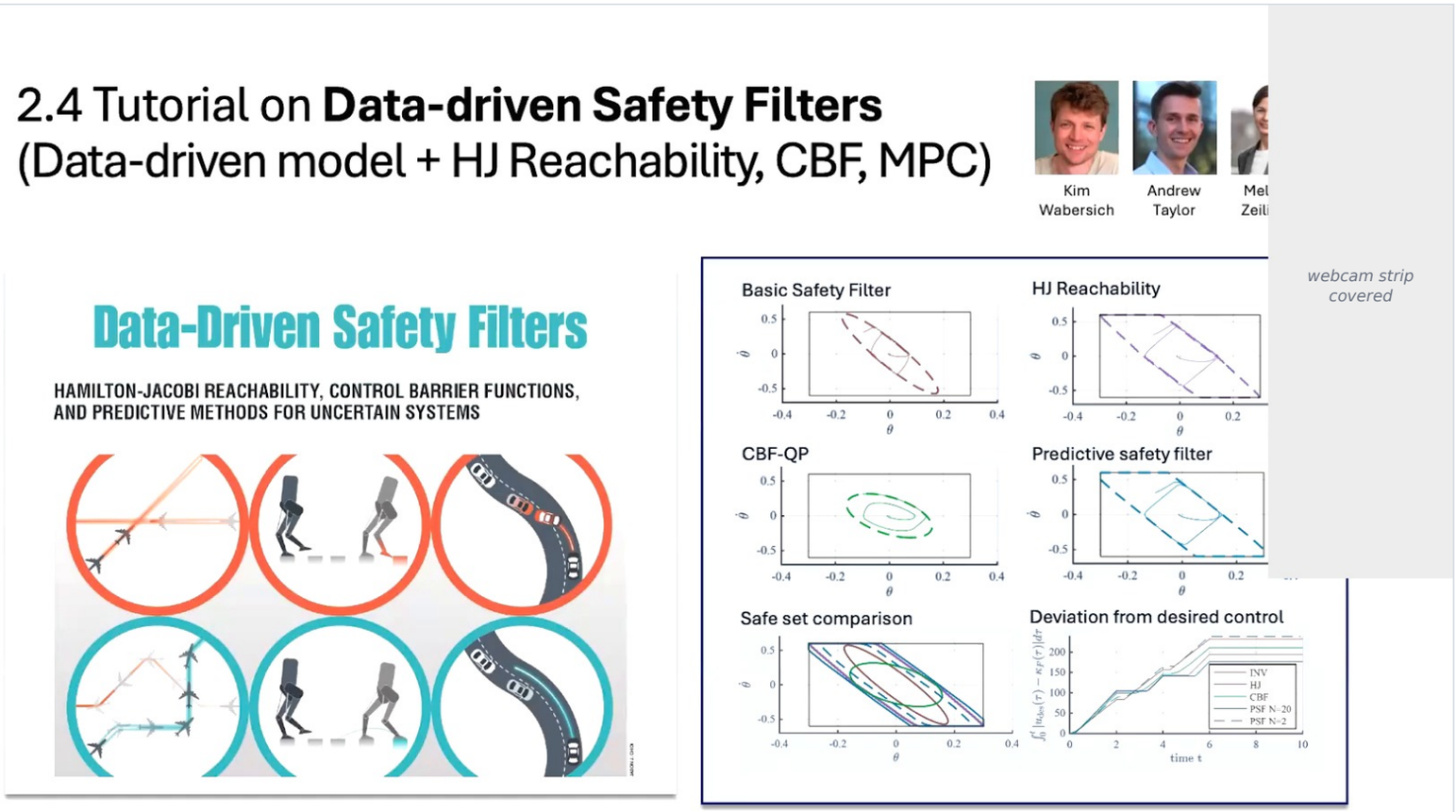
Choi*, Castaneda*, Jung* et al., RSS'20, CDC'21, T-Ro'25 (cond' accepted), Arxiv (2023)

2. INDIRECT APPROACHES FOR DATA-DRIVEN SAFETY

2.3 Learned model error + CBF

▶ 23:43

- Same idea with a control barrier function (CBF), another certificate function. Its time derivative Ḃ tells whether an input is safe.
- Learn the gap Δ between the true Ḃ and the model-based estimate, using RL or GP regression.
- Cart-pole result (plots): with the nominal model alone, the CBF filter lets the state cross the safety boundary. With the learned Δ (GP-CBF), it stays inside. (RSS'20, CDC'21, T-RO'25)



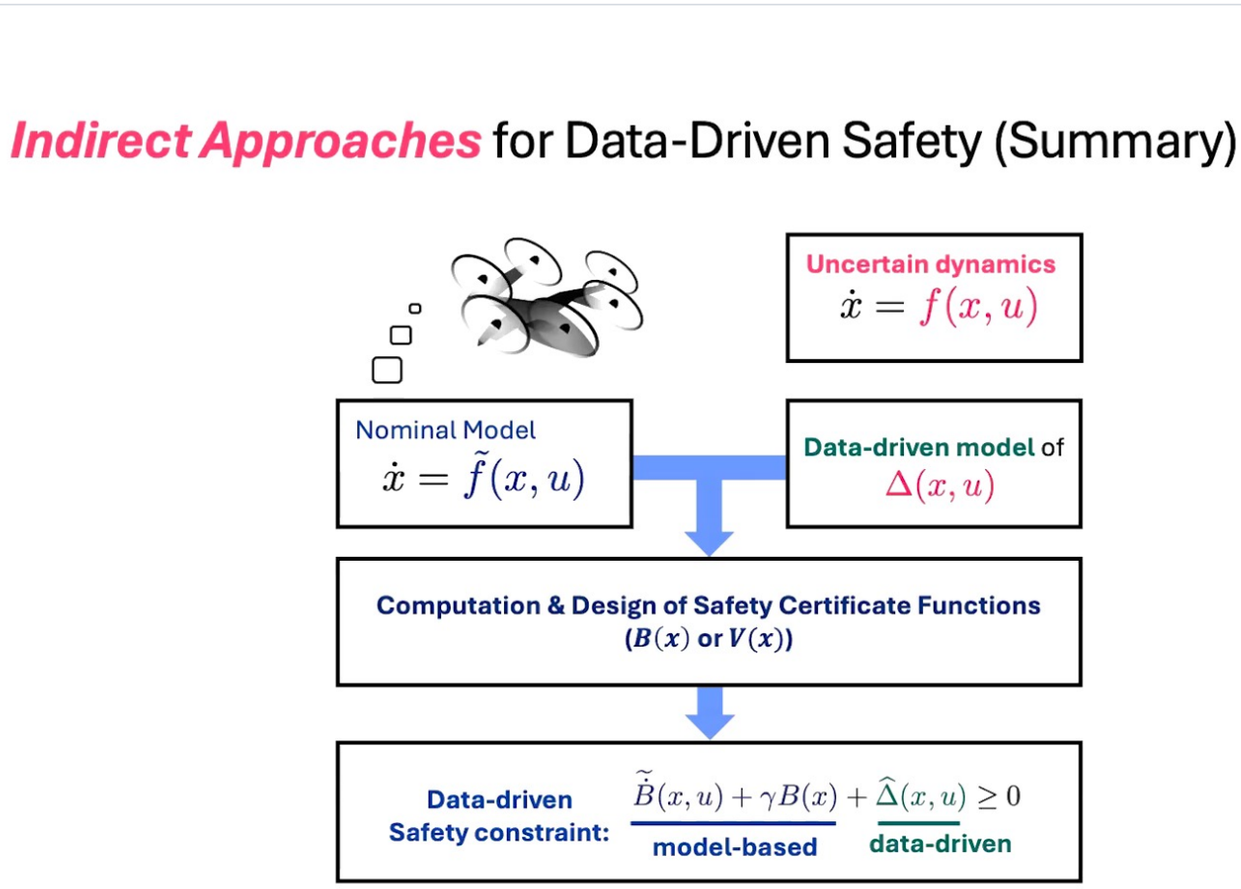
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2. INDIRECT APPROACHES FOR DATA-DRIVEN SAFETY

2.4 Survey: data-driven safety filters

▶ 24:48

- Tutorial/survey (IEEE CSM'23) with Wabersich, Taylor, Zeilinger and Ames. It covers combining data-driven methods with HJ reachability, CBFs and MPC (model predictive control).
- Common theme: a separate learned module added on top of a nominal model-based design.



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2. INDIRECT APPROACHES FOR DATA-DRIVEN SAFETY

Indirect approaches: summary

▶ 25:13

- Recipe: (1) nominal model $\tilde{f}$ from domain knowledge or system ID, (2) separate data-driven model of the error Δ, (3) build a certificate *B* or *V*, (4) a safety constraint with a model-based part and a data-driven part.
- "Indirect" because the data only enters through a separate error model.
- Lesson after ~4 years: it still takes heavy engineering (modeling, error model, filter design).



Uncertain dynamics

$$\dot{x} = f(x, u)$$

Can we ***directly*** compute / learn safe set & control input from trajectory data?

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3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Can we skip the model?

▶ 26:32

- New question: can we compute or learn the safe set and safe inputs **directly** from trajectory data?

Outline


Namhoon Cho

1. Background & Motivation
2. Indirect Approaches for Data-Driven Safety
3. **Direct Approach: Data-Driven Hamiltonian (DDH)**
4. Other works & Future research directions

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Choi*, Strong*, et al., Data-Driven Hamiltonian for Direct Construction of Safe Set from Trajectory Data, Arxiv'25 (Submitted to IEEE CDC'25)

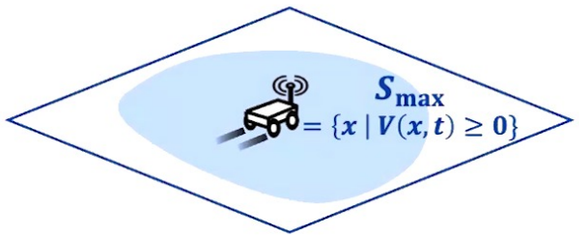
3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Part 3: Data-Driven Hamiltonian

▶ 26:39

- Most recent work, with Chris Strong and Namhoon Cho (Cranfield). arXiv 2025, submitted to IEEE CDC 2025.

Recap: Hamilton-Jacobi (HJ) Reachability



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p	: objec
$f(x,u)$	: dynar
U	: contr
x	: state

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Hamilton-Jacobi PDE:

$$0 = \min\{g(x) - V(x,t),\ -D_t V(x,t) + H(x, D_x V(x,t))\}$$

Hamiltonian:

$$H(x,p) = \max_{u \in U} p^\top f(x,u)$$

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Recap: HJ reachability

▶ 26:49

- To get *V* and the safe set, solve the HJ PDE.
- That needs H(x,p) = $\max_u p^T f(x,u)$, and *f* is exactly what we don't know.

<Notati

p : objec

$f(x,u)$: dynar

U : contr

x : state

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Q: Can we compute HJ Reachability without f (or nominal model

using trajectory data?

Hamiltonian:

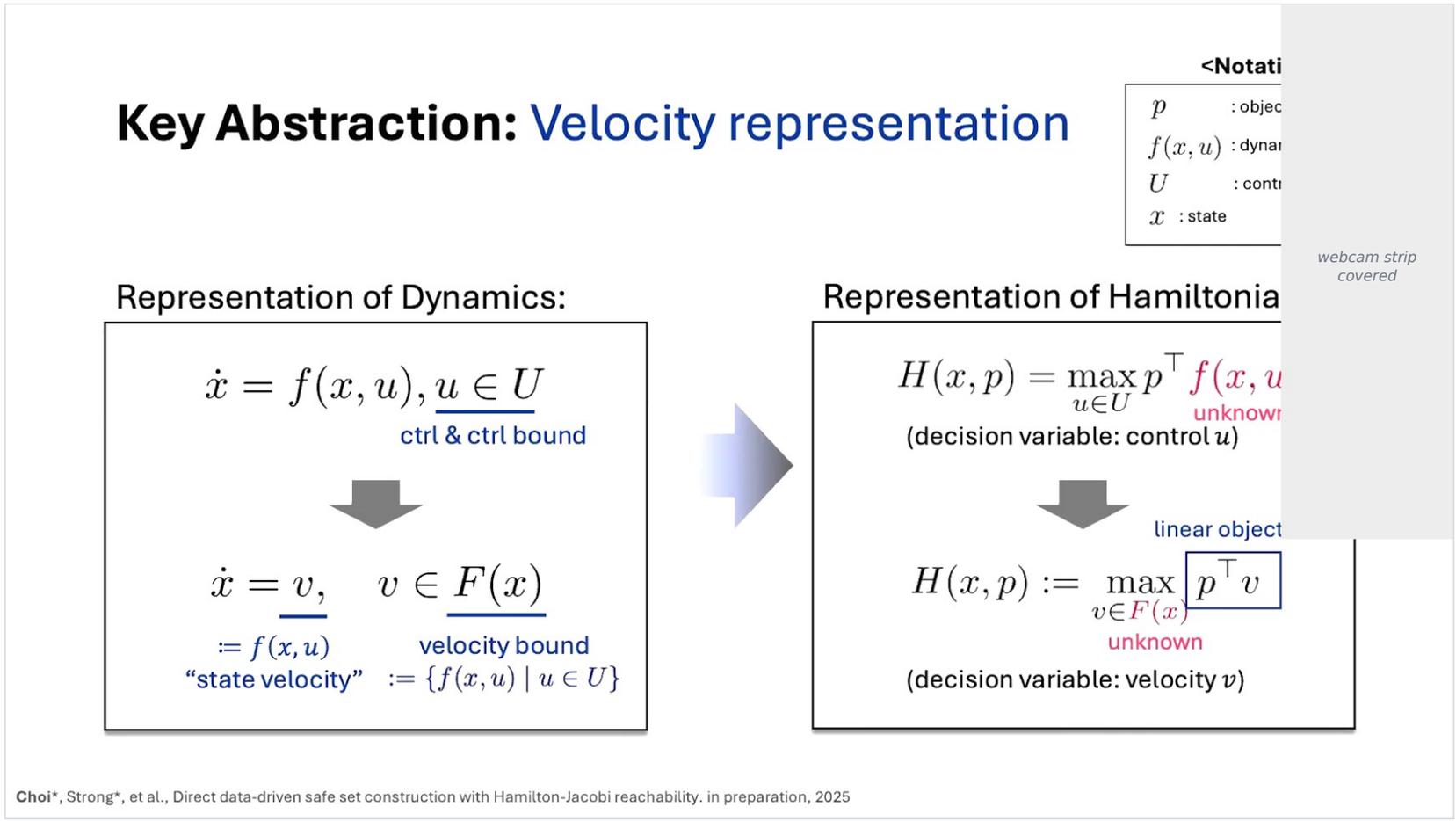
$$H(x,p) = \max_{u \in U} p^\top f(x,u)$$

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

The question

▶ 27:05

- Can we run HJ reachability without *f*, and even without a nominal model f̂, using only trajectory data?
- The key is the Hamiltonian itself.



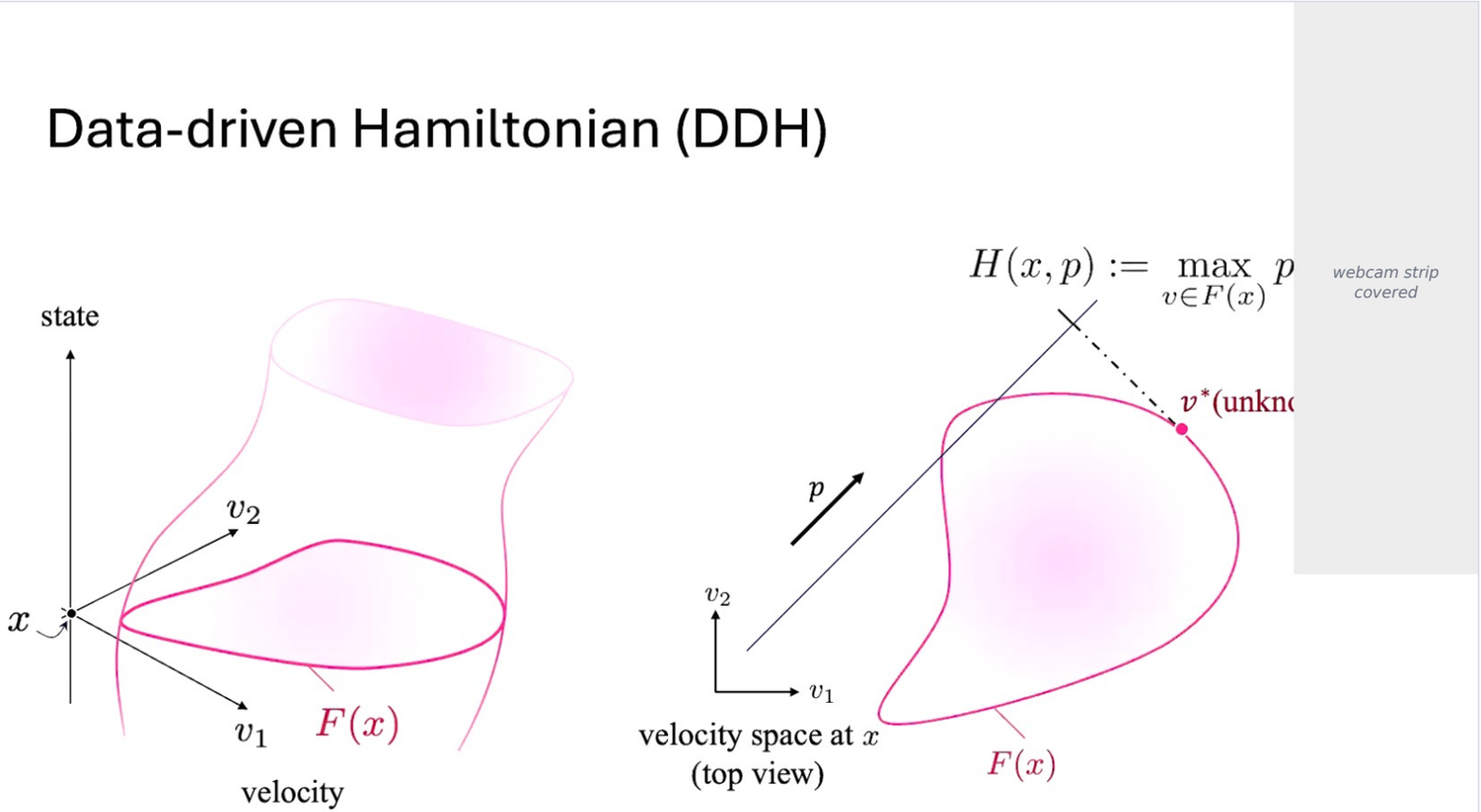
Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Key idea: think in velocities

▶ 27:19

- Rewrite $\dot{x} = f(x,u)$ as $\dot{x} = v$ with $v \in F(x)$, where $F(x)$ is the set of all velocities reachable at state x.
- Then $H(x,p) = \max$ over $v \in F(x)$ of $p^\top v$. The decision variable changes from the control u to the velocity v. The objective is now linear in v, and the unknown has moved into the set F(x).
- Example: for p = (1, 0), *H* is the largest first component of any reachable velocity.



Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

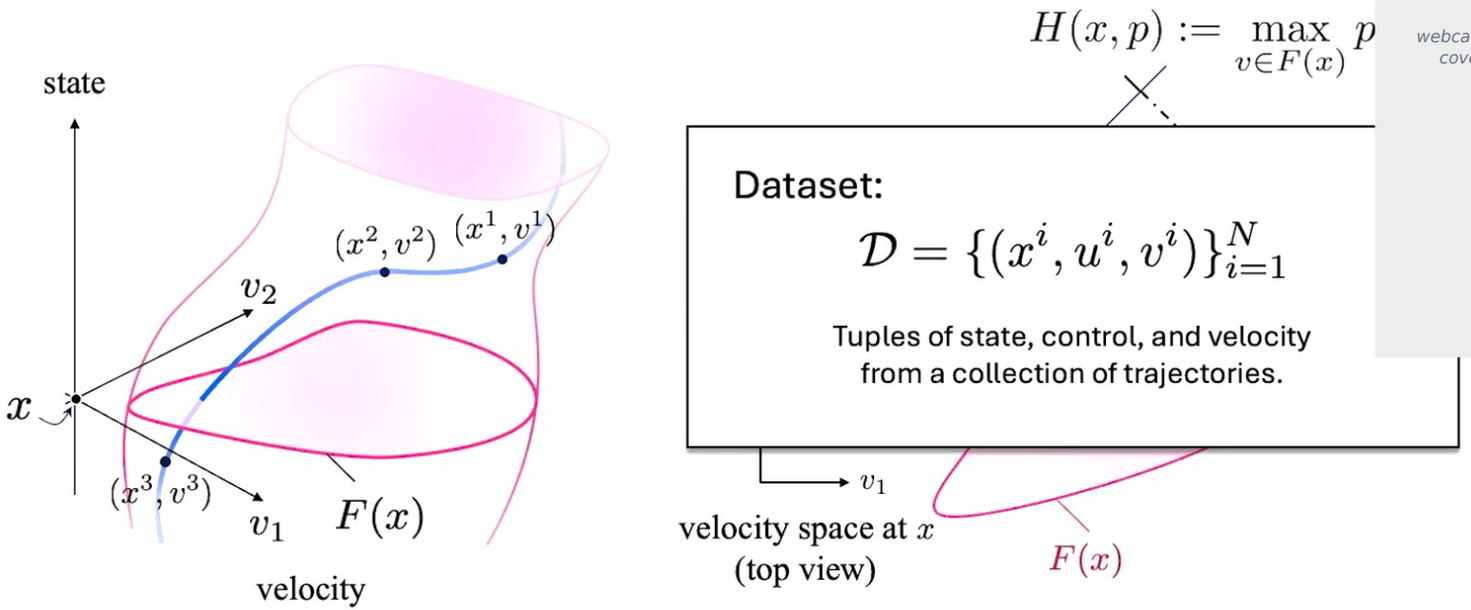
3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Picture: the unknown velocity set

▶ 28:31

- At each state x there is an unknown set F(x) of possible velocities (shown from above on the right).
- *H*(x,p) is reached at the point v* of F(x) farthest along p. We can't find v* without knowing F.

Data-driven Hamiltonian (DDH)



$$H(x,p) := \max_{v \in F(x)} p$$

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Dataset:

$$\mathcal{D} = \{(x^i, u^i, v^i)\}_{i=1}^{N}$$

Tuples of state, control, and velocity from a collection of trajectories.

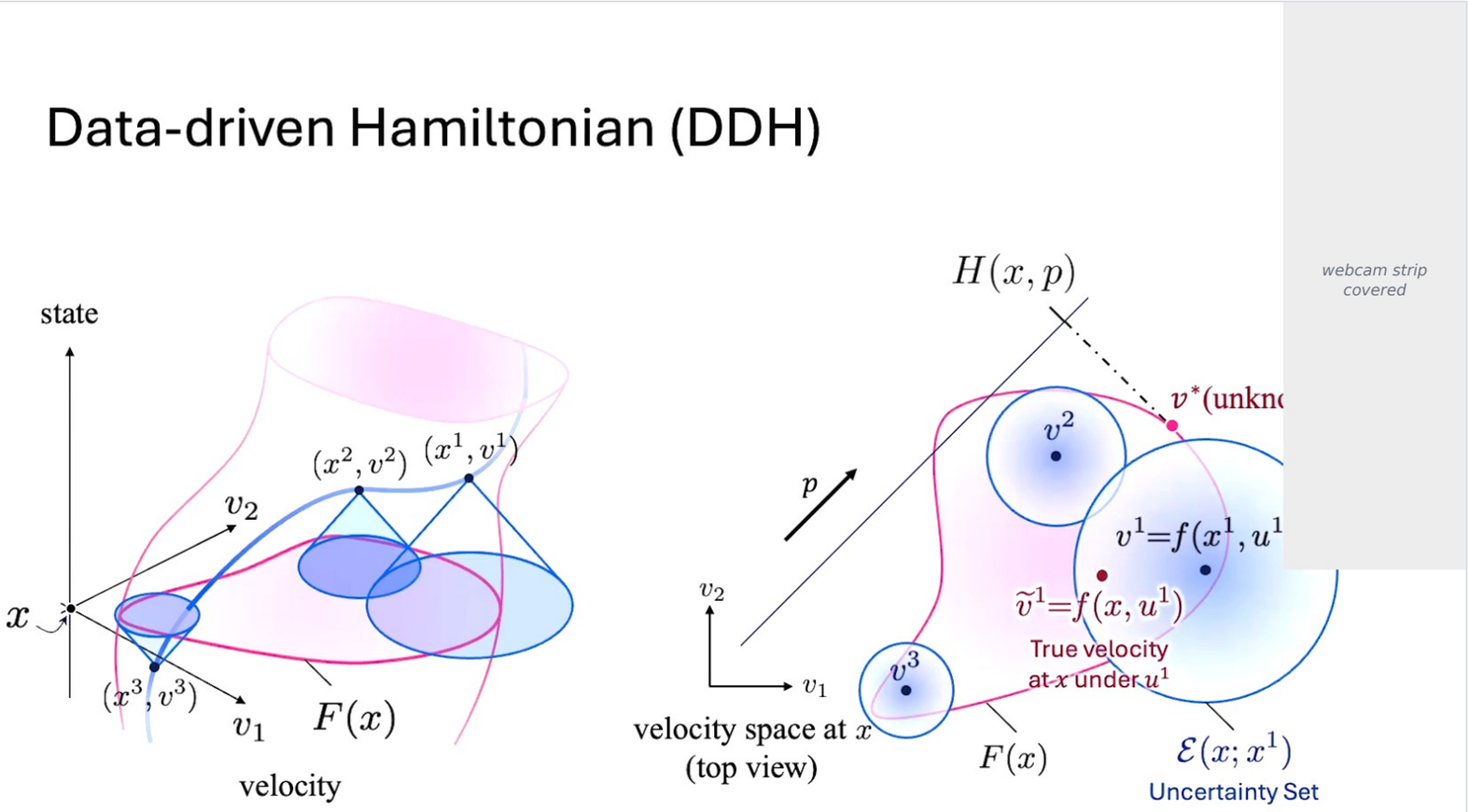
Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

The dataset

▶ 29:18

- Data D = {(x^i, u^i, v^i)}, i = 1…N: tuples of state, control and measured velocity from many trajectories.
- Each trajectory is a curve through state-velocity space.



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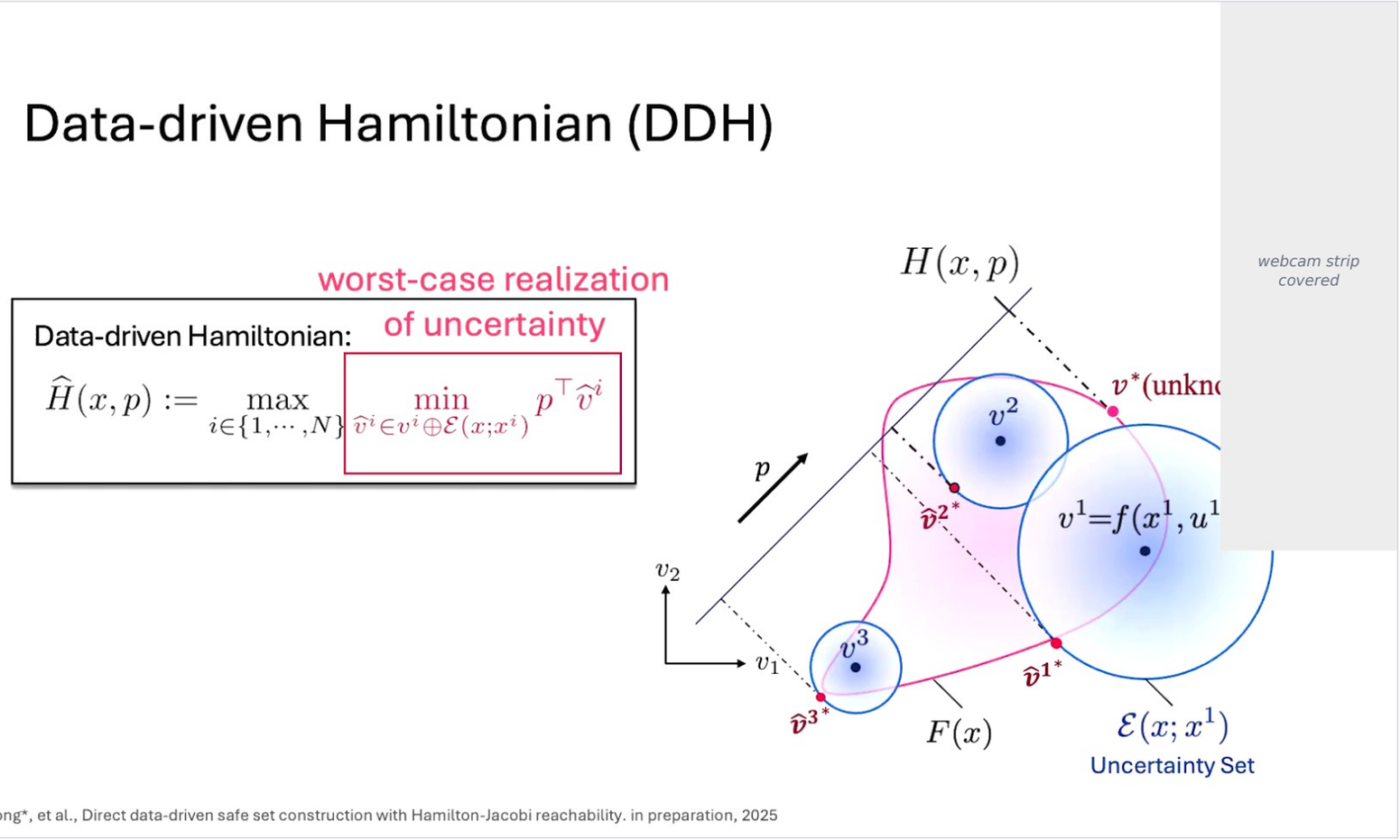
Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

From data points to uncertainty balls

▶ 29:35

- Data were measured at other states, not at the current state x.
- Using smoothness of the dynamics, move each point to x with an uncertainty ball $E(x; x^i)$. Points farther from x get bigger balls.
- Guarantee: the true velocity $f(x, u^i)$ lies inside its ball.



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3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

DDH: worst case inside each ball

▶ 30:30

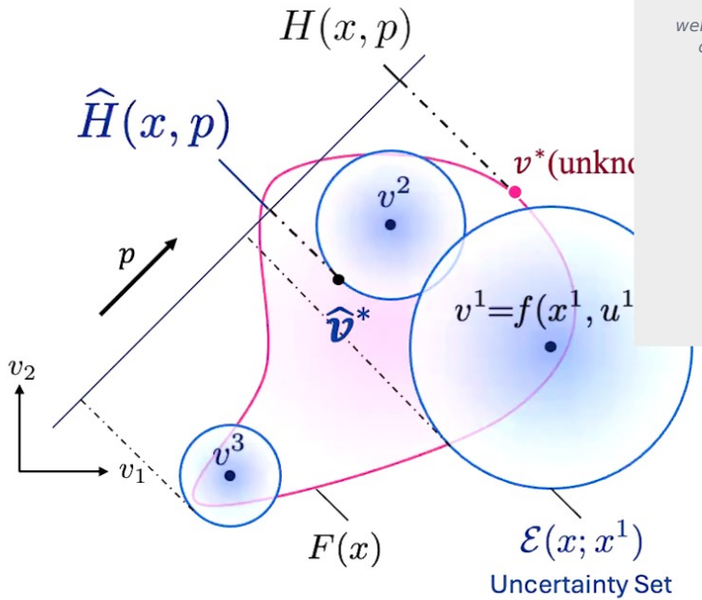
- Data-Driven Hamiltonian: $\hat{H}(x,p)$ = max over i of [min over $\tilde{v}$ in ball i of $p^T\tilde{v}$].
- The inner min takes the worst-case velocity in each ball along direction p (the starred points on the slide).

Data-driven Hamiltonian (DDH)

Data-driven Hamiltonian:

$$\widehat{H}(x,p) := \max_{i\in\{1,\cdots,N\}} \min_{\widehat{v}^i \in v^i \oplus \mathcal{E}(x;x^i)} p^\top \widehat{v}^i$$

best guess among data



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3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

DDH: best guess among the data

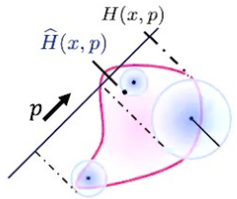
▶ 30:53

- The outer max picks the best of those worst cases over all data points.
- It is a convex problem with a closed-form solution, whatever the true dynamics look like.
- Because it always takes the worst case, $\hat{H} \leq H$: it never overestimates.

Guarantees with DDH

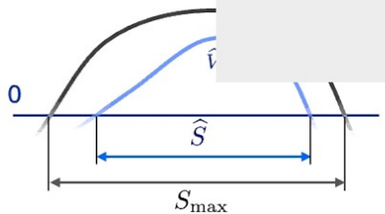
1. DDH is a guaranteed under-approximation.

$$\widehat{H}(x,p) \leq H(x,p)$$



2. Data-driven value function is a guaranteed under-approximation.

$$\widehat{V}(x,t) \leq V(x,t)$$



3. Data-driven safe set is a guaranteed inner-approximation.

$$\widehat{S} = \{x \mid \widehat{V}(x,t) \geq 0\} \quad \subseteq \quad S_{\max} = \{x \mid V(x,t) \geq 0\}$$

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3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Guarantees with DDH

▶ 31:30

- $\hat{H}(x,p) \leq H(x,p)$, so the data-driven value function is $\leq$ the true V.
- So the data-driven safe set $\hat{S} \subseteq S_{max}$: an **inner approximation**.
- This is the right direction for safety. Any state the data calls safe is truly safe.

Direct Approach for Data-Driven Safety (Summary)

1. **Data-driven Hamiltonian (DDH)** provides a new paradigm of integrating model-based analysis & data-driven approaches for safety.

2. The key idea is circumventing explicit modeling of dynamics with **effective abstraction** (velocity representation).

3. Guarantees **inner-approximation** of the safe set.

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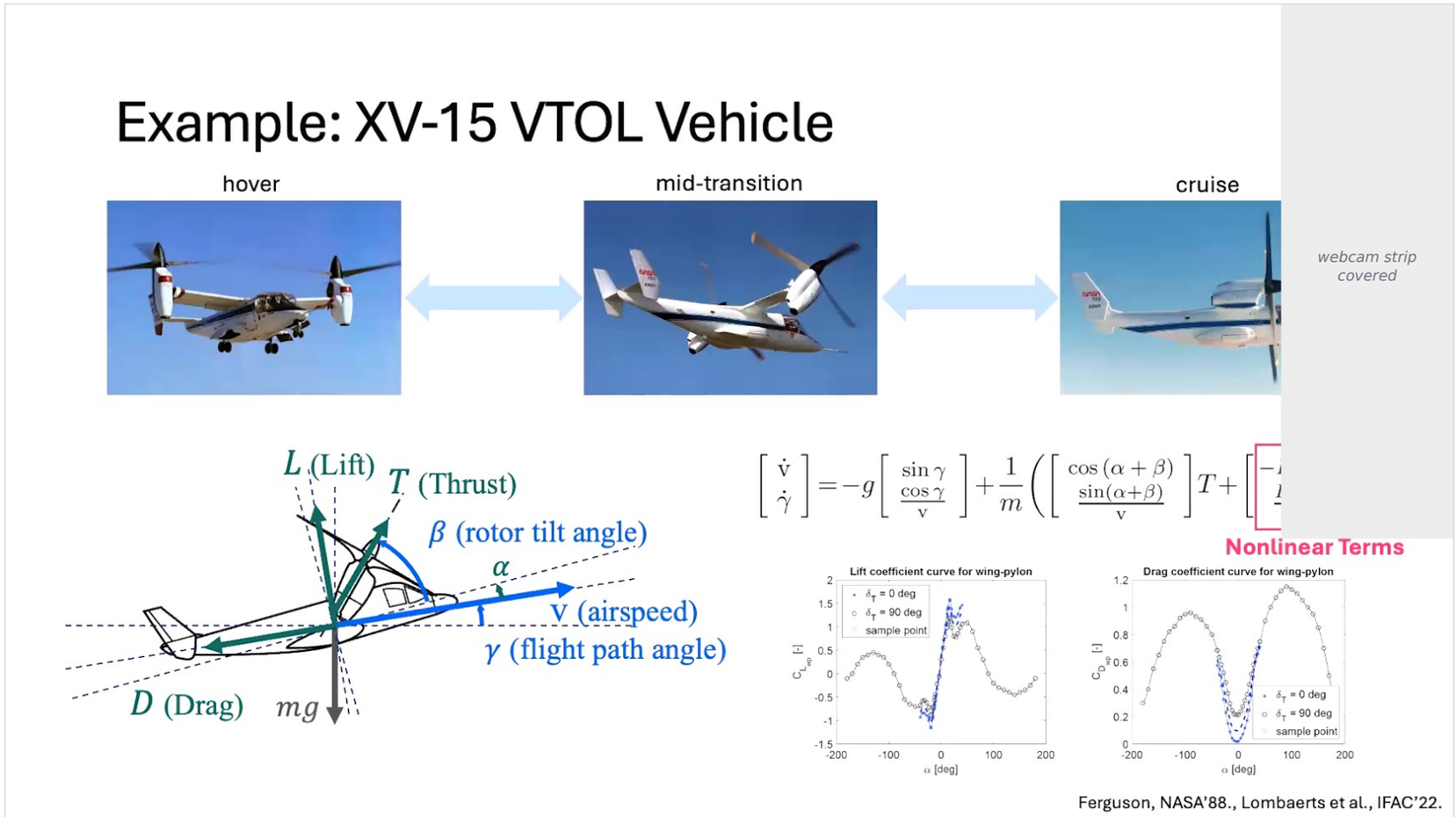
Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Direct approach: summary

▶ 32:06

- DDH feeds data straight into model-based HJ analysis, with no explicit dynamics model.
- Key idea: think in terms of velocities, which removes the specific form of *f*.
- Guaranteed inner approximation of the true safe set.



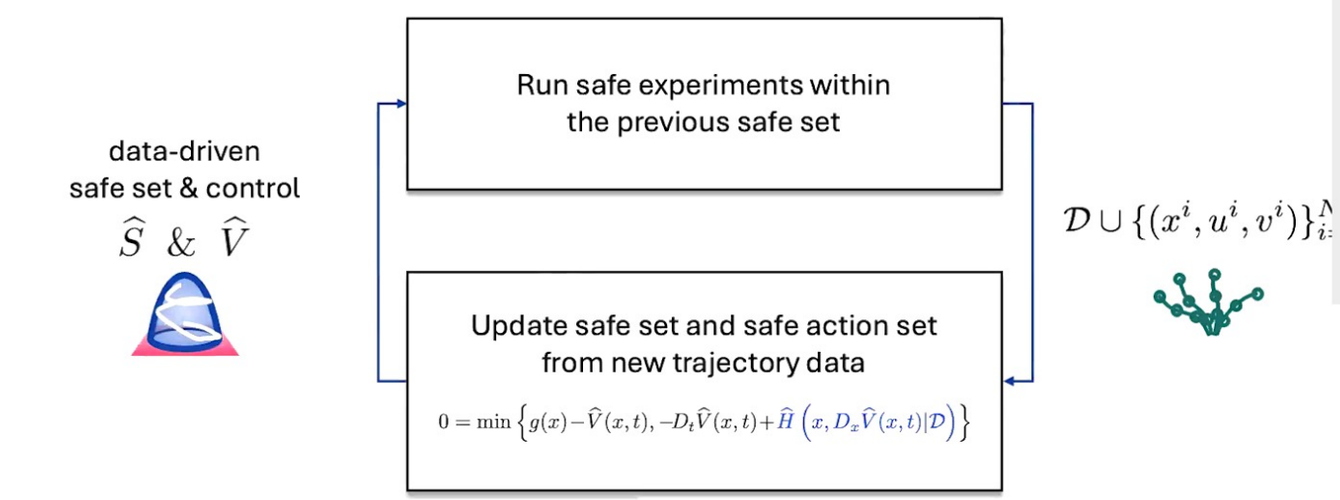
3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Example: XV-15 tiltrotor

▶ 32:53 · revisited at 42:29

- XV-15: an ancestor of today's air taxis, and one of the few tiltrotors with a well-documented true model. That lets them check the data-driven result against ground truth.
- Longitudinal model with 3 states: airspeed V, flight-path angle γ, rotor tilt β (90° = hover, 0° = cruise). 3 inputs: thrust, angle of attack, tilt rate dβ/dt. Lift and drag are strongly nonlinear.
- Known parts (here the dβ/dt dynamics) can be plugged in as a model. DDH is modular.

Q&A: *The audience questioned treating angle of attack as a direct input. He agreed it is a simplified (reduced-order) model.*



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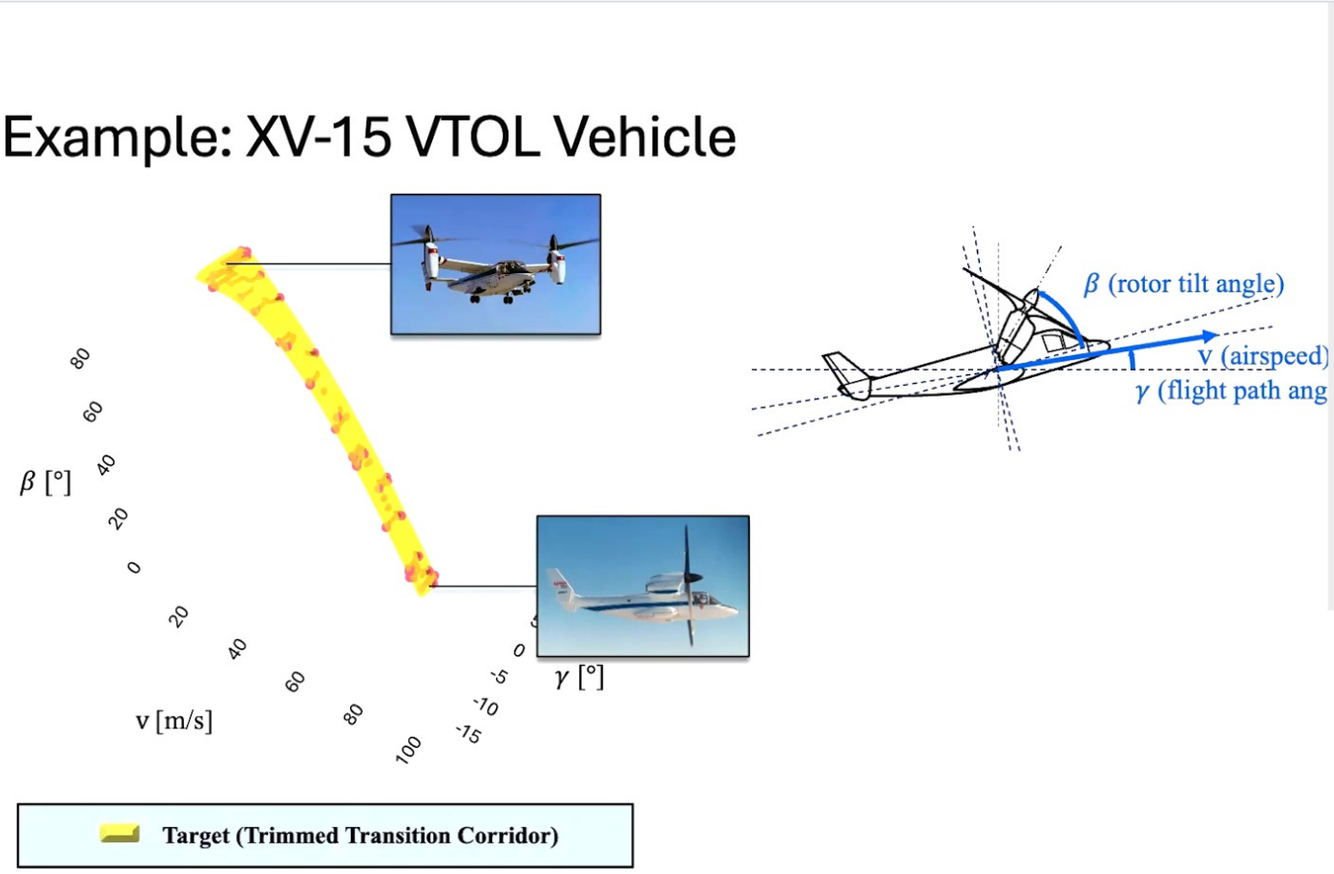
Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Growing the safe set with safe data collection

▶ 35:29

- Collecting the data must also be safe.
- Loop: run experiments only inside the current safe set, add the data, recompute the safe set and filter with DDH, then explore the larger set.
- Start from a small, conservative set and repeat until the set is large enough.



Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

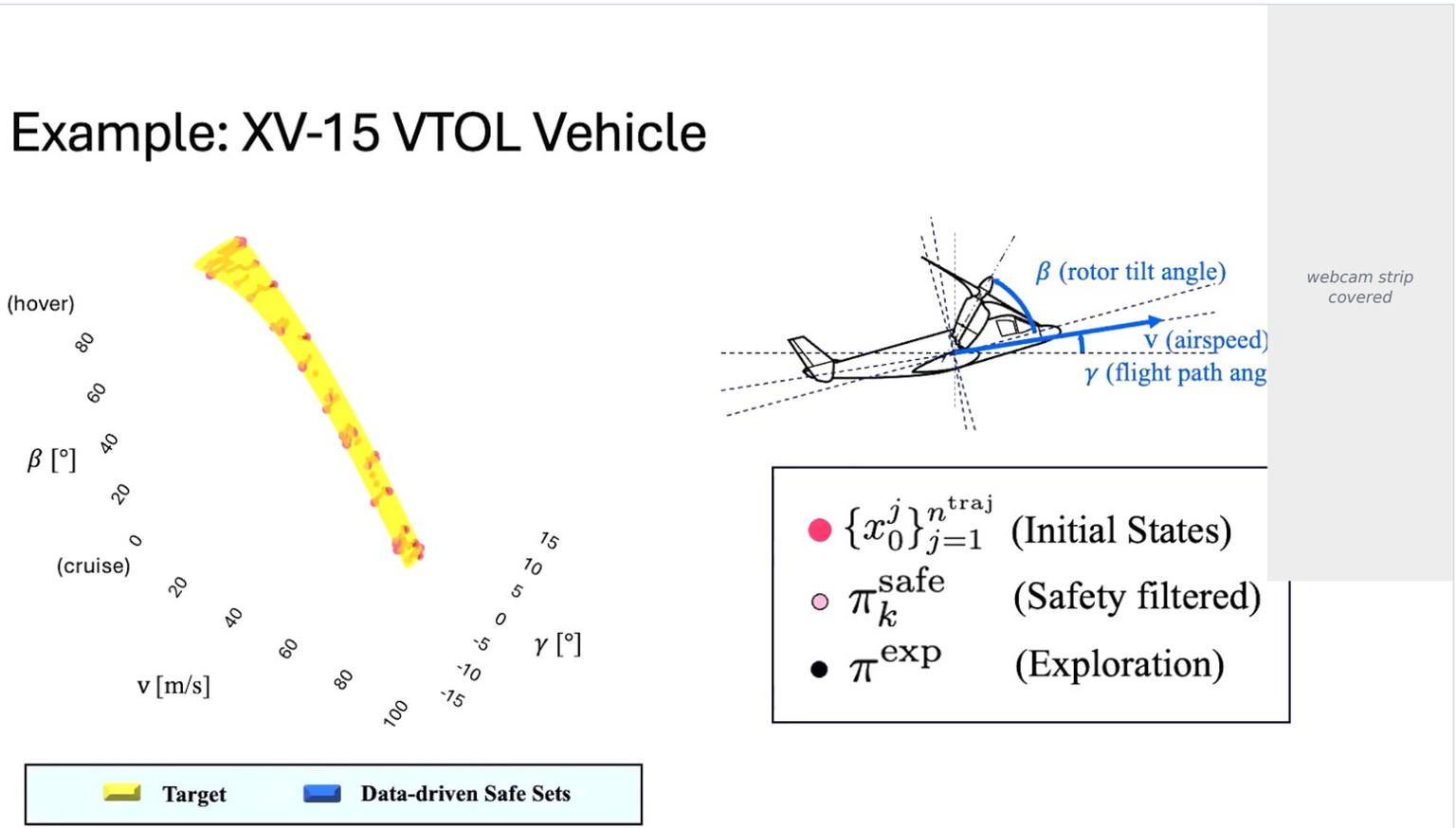
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3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Target: the transition corridor

▶ 36:10

- Yellow target = the trimmed transition corridor that links hover and cruise.
- If the vehicle can always get back to this corridor, the mode transition can be done safely.



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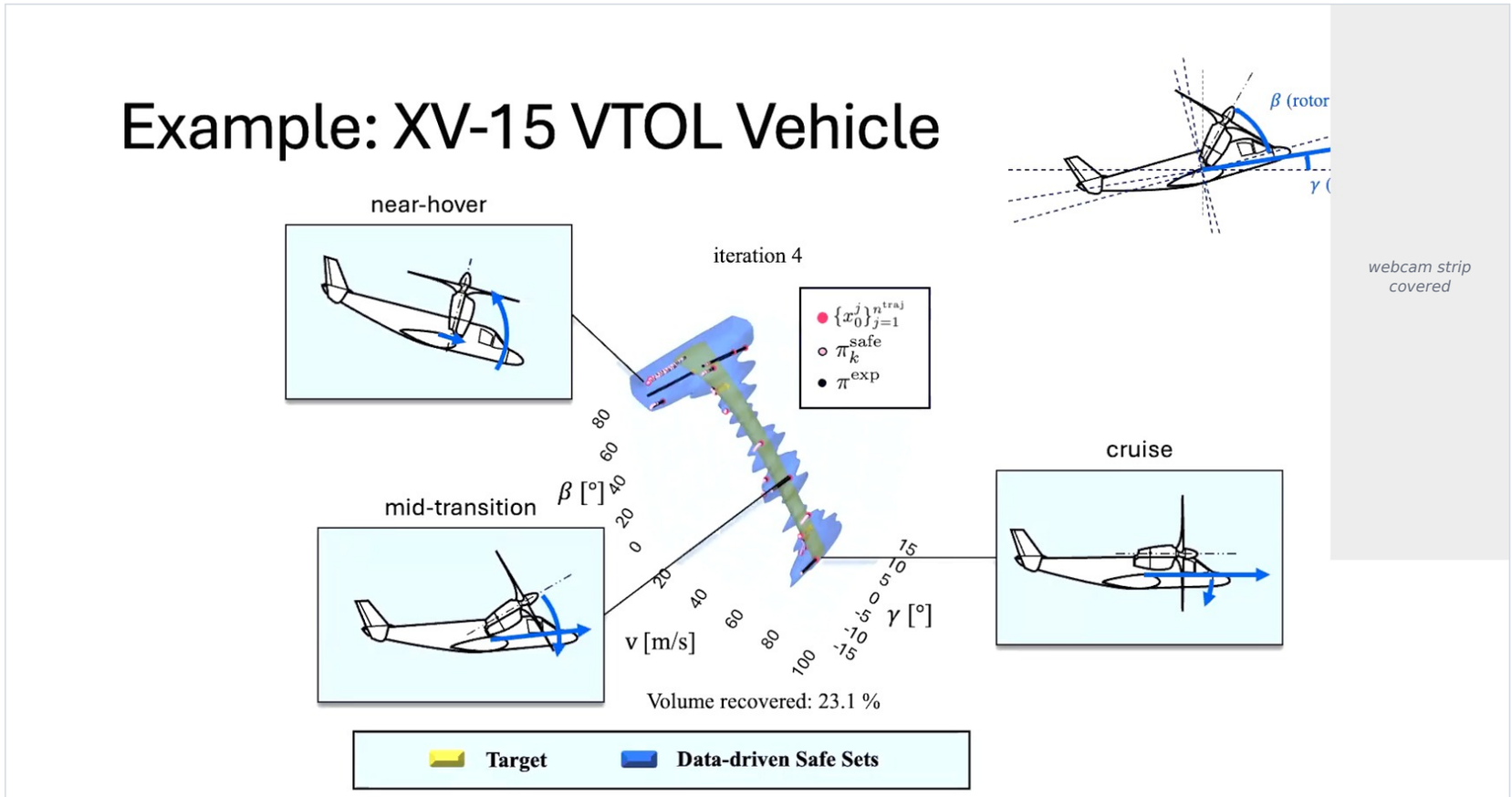
Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Experiments in each round

▶ 36:31 · revisited at 42:17

- Each round runs 20 trajectories (red dots = start states). An exploration policy flies them, and a DDH-based safety filter steps in when a trajectory is about to leave the safe set.
- After each round, the new data update the safe set. Safe-set volume grows from 13.6% of the true set (round 0) to 23.1% (round 4).



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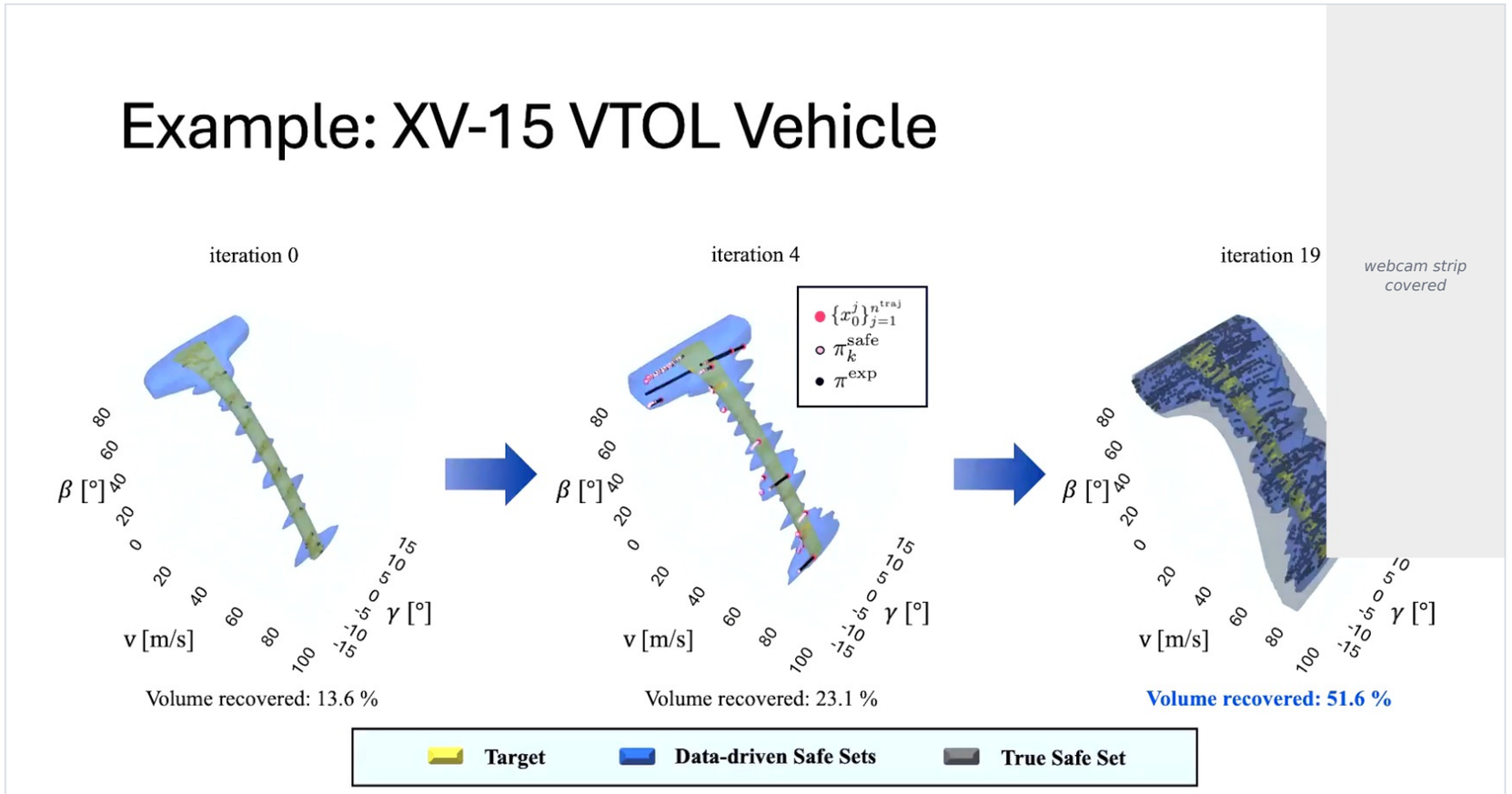
Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Sample trajectories (round 4)

▶ 37:21 · revisited at 43:37

- Runs start near hover, mid-transition and in cruise, all inside the previous safe set. All stay safe thanks to the data-driven filter.
- Trajectories look mostly horizontal because the tilt rate dβ/dt is slow in this model.



Choi*, Strong*, et al., Direct data-driven safe set construction with Hamilton-Jacobi reachability. in preparation, 2025

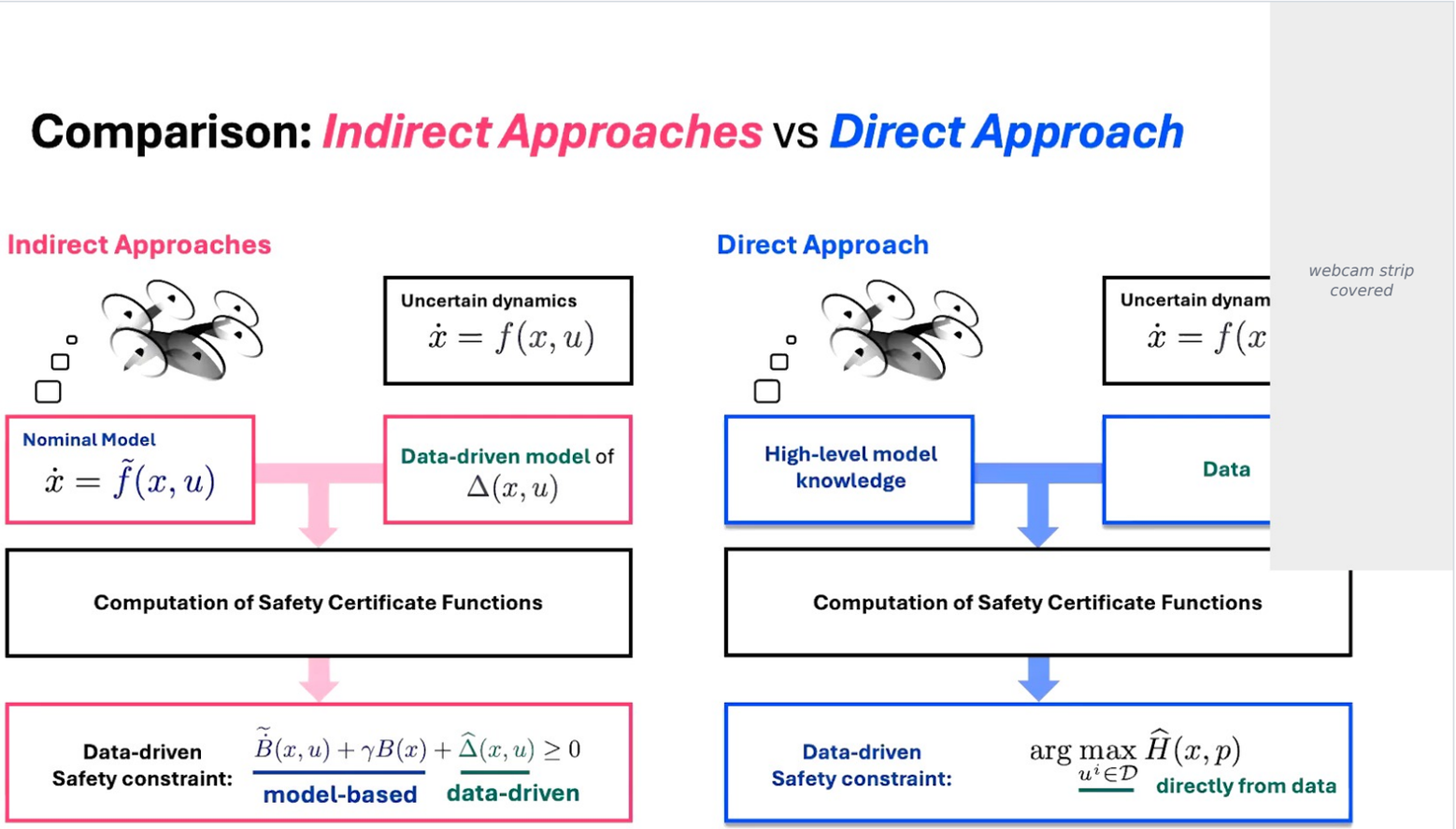
3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Result: safe set over the rounds

▶ 37:49 · revisited at 43:53

- Recovered volume: 13.6% → 23.1% → **51.6%** of the true safe set after 19 rounds. More rounds should recover more.
- Cost: the same as standard HJ, plus a part that grows linearly with the number of data points. About 30 min per round, vs 30–60 min for the ground truth (which needs discretized inputs because it is not control-affine).
- No interpolation between trajectories is needed. DDH is used inside the reachability solver.

Q&A: *The audience asked whether results on this reduced-order model carry over to a real aircraft. He acknowledged the gap and said it is future work.*



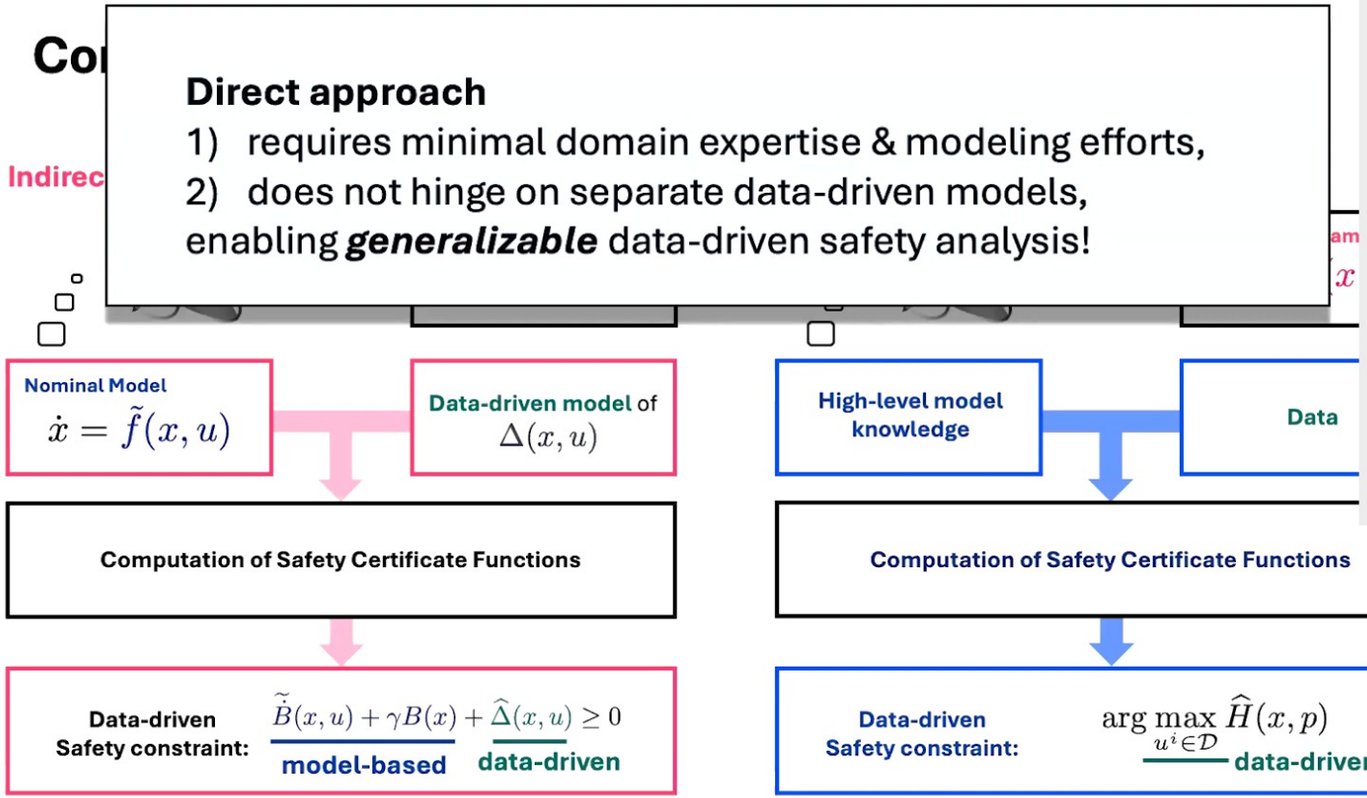
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3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Comparison: indirect vs direct

▶ 44:34

- **Indirect:** nominal model $\tilde{f}$ + a separate data-driven model of the error Δ, then certificate functions. The safety constraint has a model-based part and a data-driven part.
- **Direct:** high-level model knowledge + data, then certificate functions. The safety constraint (argmax of $\hat{H}$ over the data) comes directly from data.



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3. DIRECT APPROACH: DATA-DRIVEN HAMILTONIAN

Why the direct approach generalizes

▶ 44:39

- The direct approach (1) needs minimal domain expertise and modeling effort, and (2) does not depend on separate data-driven models.
- So it enables data-driven safety analysis that generalizes across systems.

Outline

1. Background & Motivation
2. Indirect Approaches for Data-Driven Safety
3. Direct Approach: Data-Driven Hamiltonian (DDH)
4. **Other works & Future research directions**

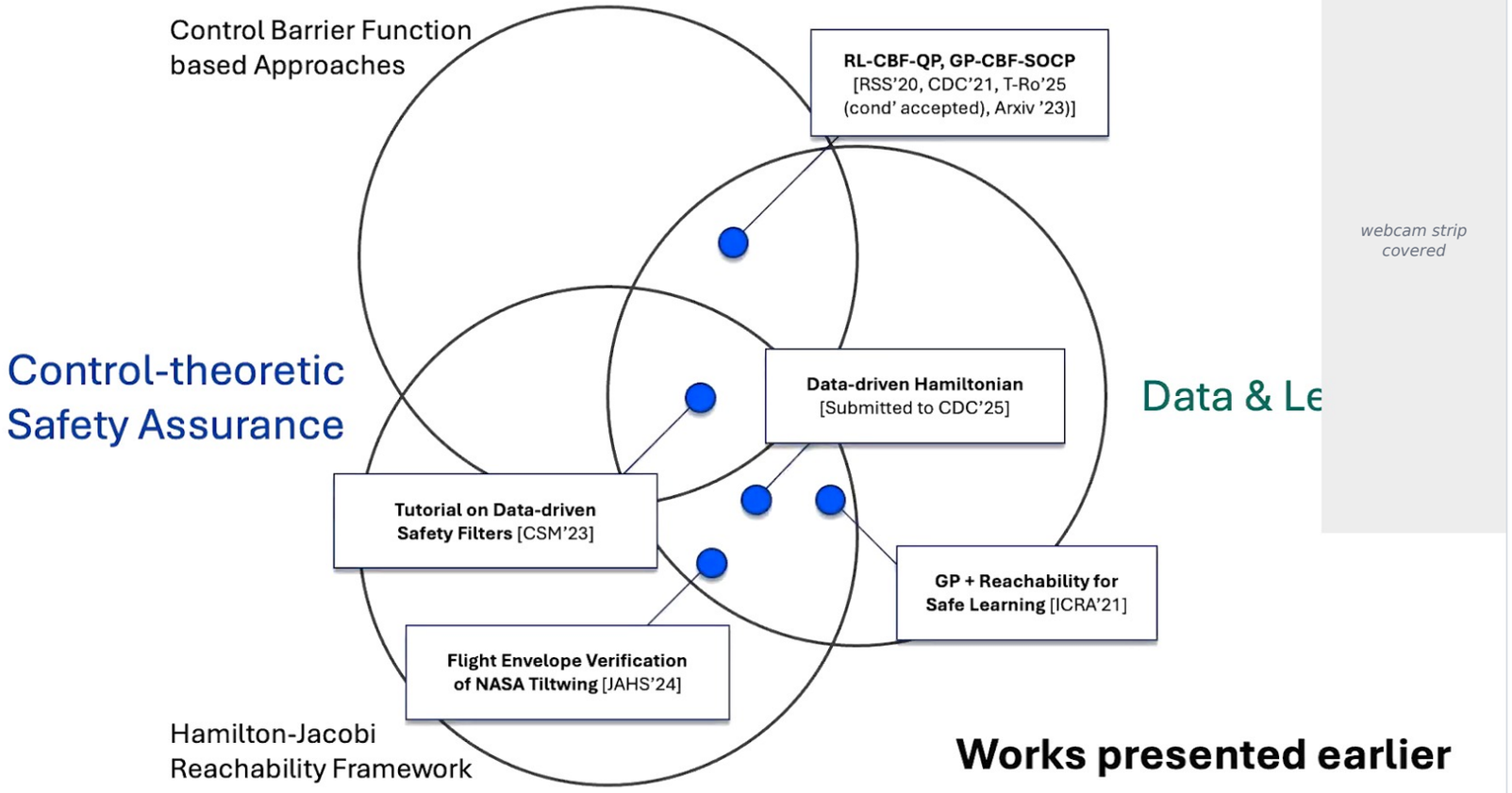
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4. OTHER WORKS & FUTURE DIRECTIONS

Part 4: Other works & future directions

▶ 45:03

- Wrap-up: other projects and where the research is heading.



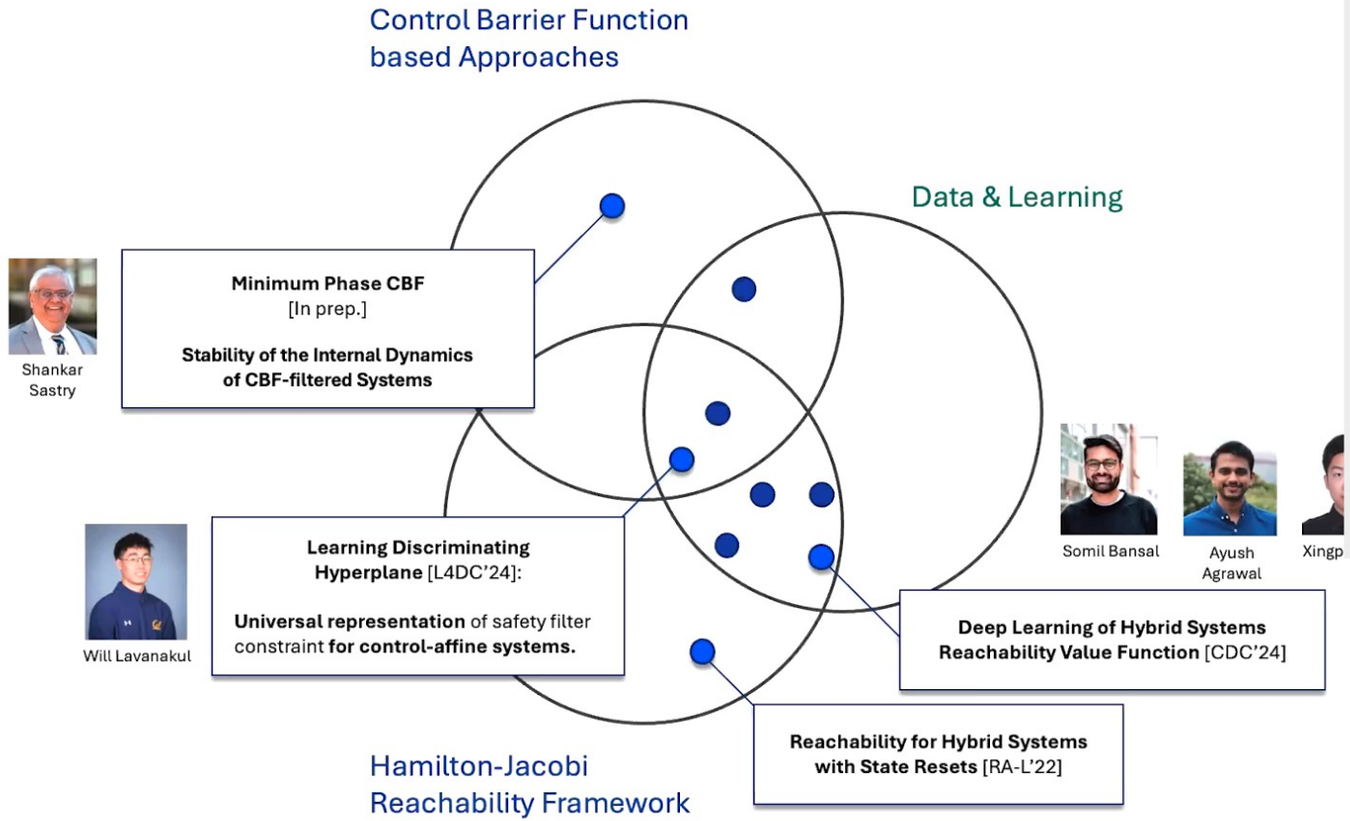
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4. OTHER WORKS & FUTURE DIRECTIONS

Map of his research

▶ 45:09

- His papers sit where three areas overlap: CBF methods, HJ reachability, and data & learning.
- Covered today: RL/GP-based CBF filters, GP + reachability (ICRA'21), NASA Tiltwing envelope (JAHS'24), the safety-filter tutorial (CSM'23), and the Data-driven Hamiltonian (submitted to CDC'25).



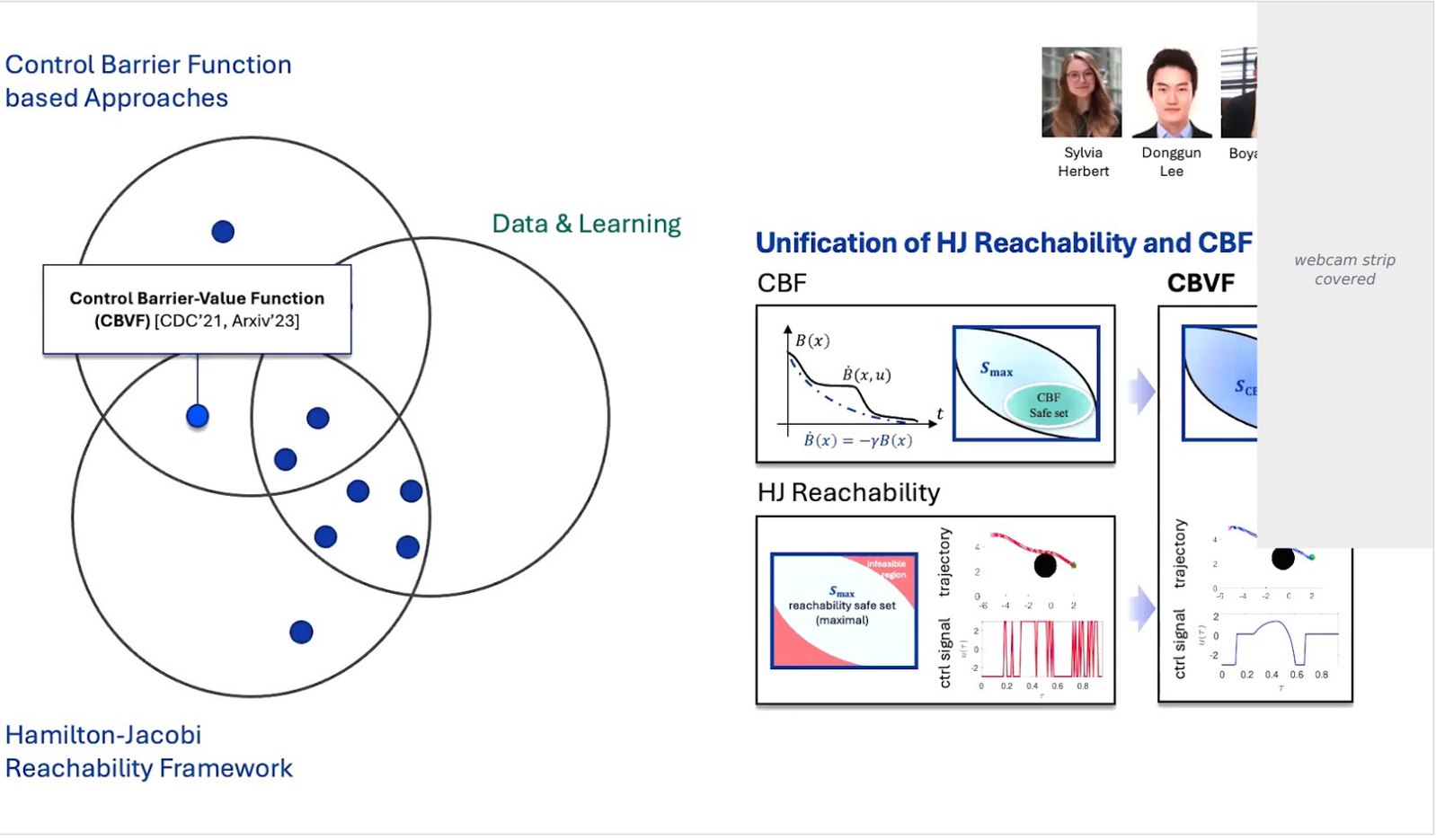
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4. OTHER WORKS & FUTURE DIRECTIONS

Other works

▶ 45:25

- Minimum-phase CBF: keeping internal dynamics stable under CBF filtering (in prep.).
- Learning discriminating hyperplanes (L4DC'24): a cheap, general form of safety-filter constraint for control-affine systems.
- Reachability for hybrid systems with state resets, e.g. legged robots (RA-L'22), and deep learning of their value functions (CDC'24).



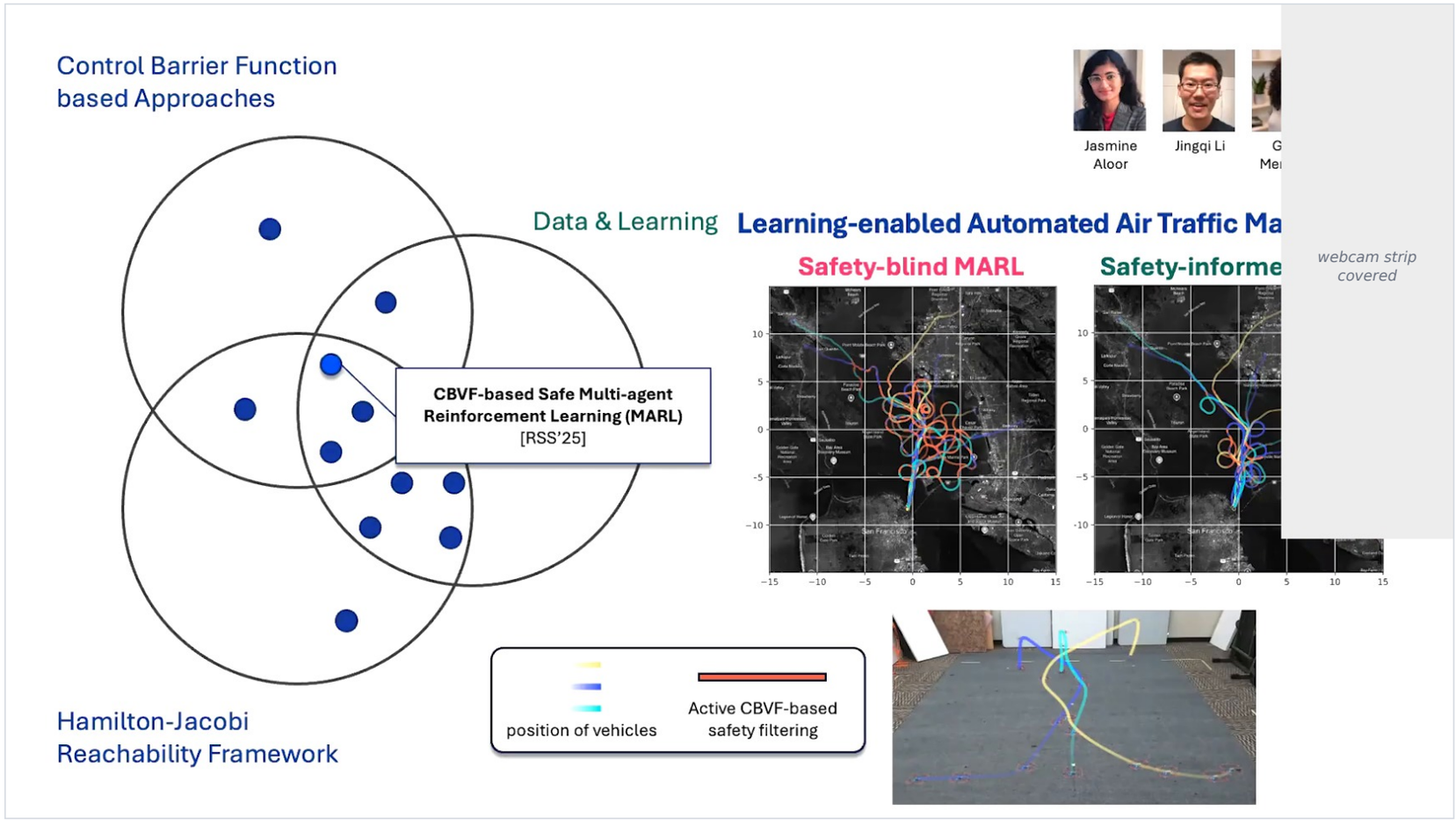
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4. OTHER WORKS & FUTURE DIRECTIONS

Control Barrier-Value Function (CBVF)

▶ 45:45

- Unifies HJ reachability and CBFs (CDC'21, arXiv'23).
- HJ gives the largest safe set but an on/off filter. A CBF gives smooth filtering but is hard to find. CBVF aims for the benefits of both.

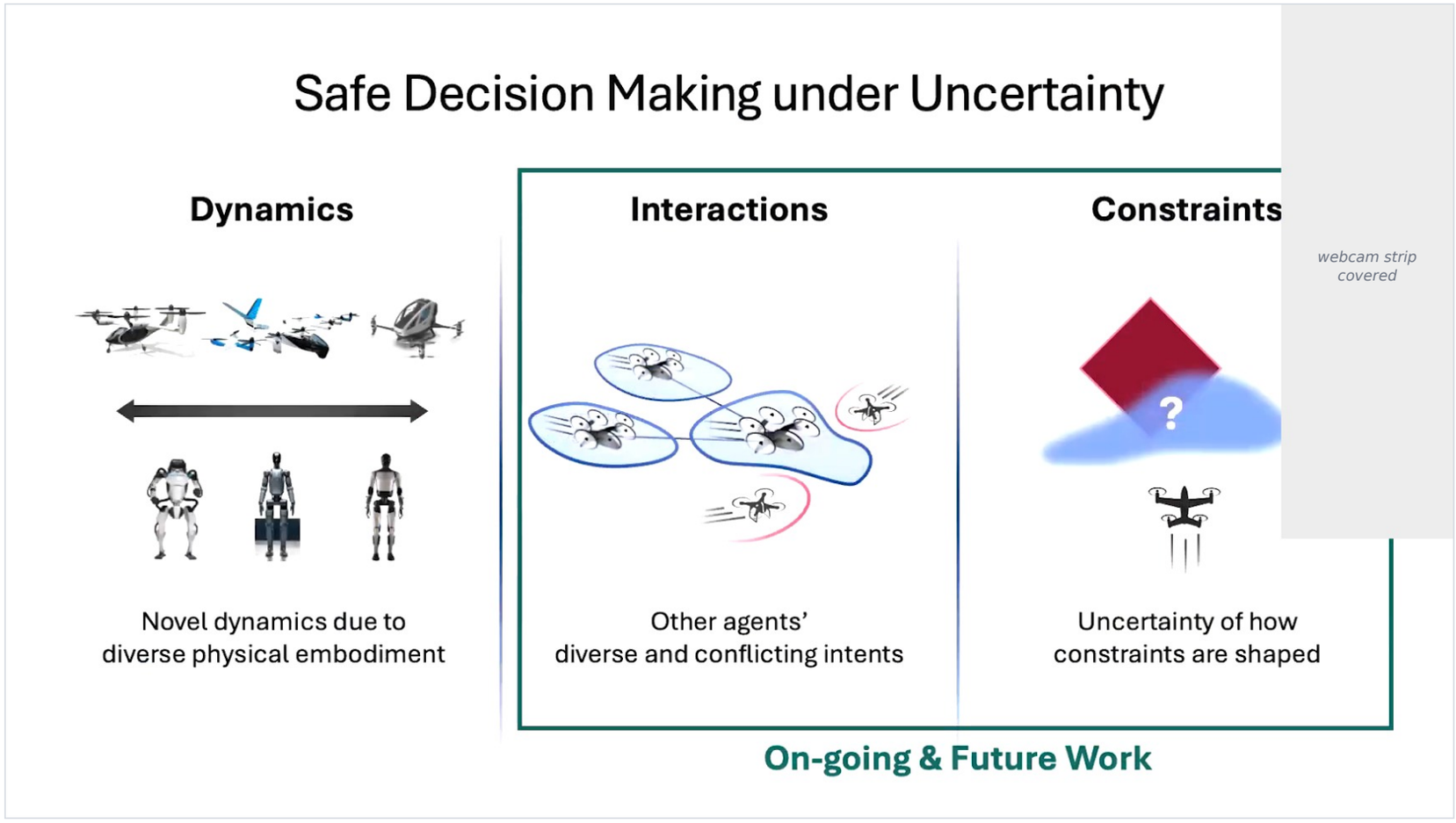


4. OTHER WORKS & FUTURE DIRECTIONS

Safe multi-agent RL for air traffic

▶ 46:04

- A CBVF-based safety filter keeps a multi-agent reinforcement learning (MARL) policy safe for automated air traffic management (RSS'25).
- Using the safety analysis to shape the RL reward gives a more efficient policy than a safety-blind baseline. Also tested on hardware.



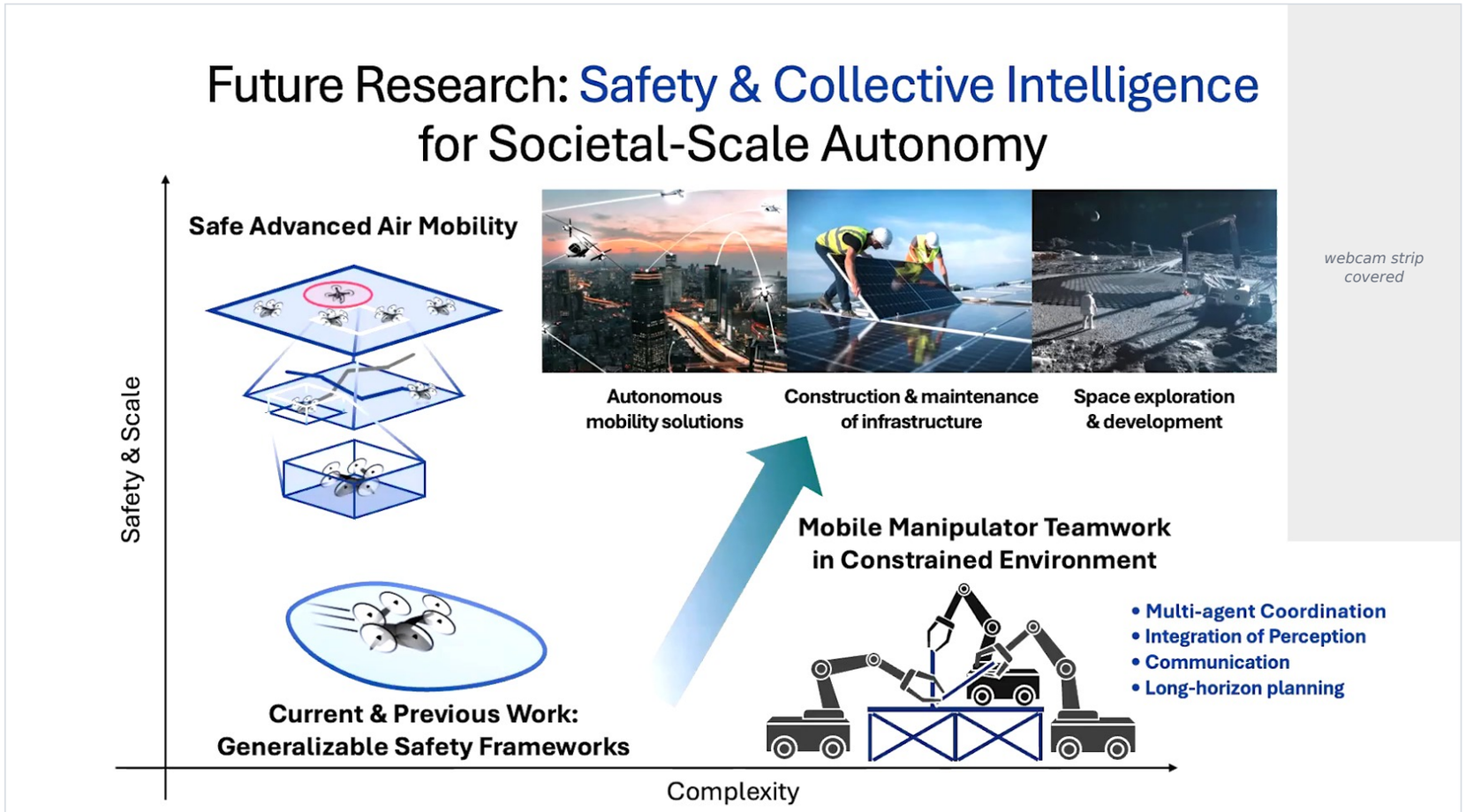
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4. OTHER WORKS & FUTURE DIRECTIONS

Back to the big picture

▶ 46:44

- Recap of the three sources of uncertainty: dynamics, interactions and constraints.
- The PhD tackled uncertain dynamics. Interactions and constraints are on-going and future work.



4. OTHER WORKS & FUTURE DIRECTIONS

Future research

▶ 46:47

- Two directions: **scale** (safe advanced air mobility, with many vehicles and strict safety standards) and **complexity** (teams of mobile manipulators doing construction).
- Open problems: multi-agent coordination, perception, communication, long-horizon planning.
- Long-term goal: safe, collectively intelligent autonomy at societal scale.

Research Advisors



Koushil Sreenath



Claire Tomlin

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ACKNOWLEDGEMENTS

Research advisors

▶ 47:55

- Thanks to Koushil Sreenath and Claire Tomlin.
- From Koushil: work toward concrete milestones (e.g. build a proof of concept within a month). From Claire: start from deep, abstract thinking for long-term impact. He found the two styles complementary.

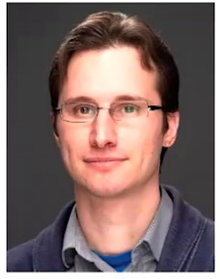
PhD Committee Members



Shankar Sastry



Mark Mueller



Sergey Levine

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ACKNOWLEDGEMENTS

PhD committee

▶ 49:51

- Shankar Sastry (he was a GSI for the nonlinear systems course), Mark Mueller (estimation), Sergey Levine (learning perspective).

Other Mentors (Academia)



Sylvia Herbert
(UCSD)



Donggun Lee
(NCSU)



Somil Bansal
(Stanford)



Namhoon Cho
(Cranfield)

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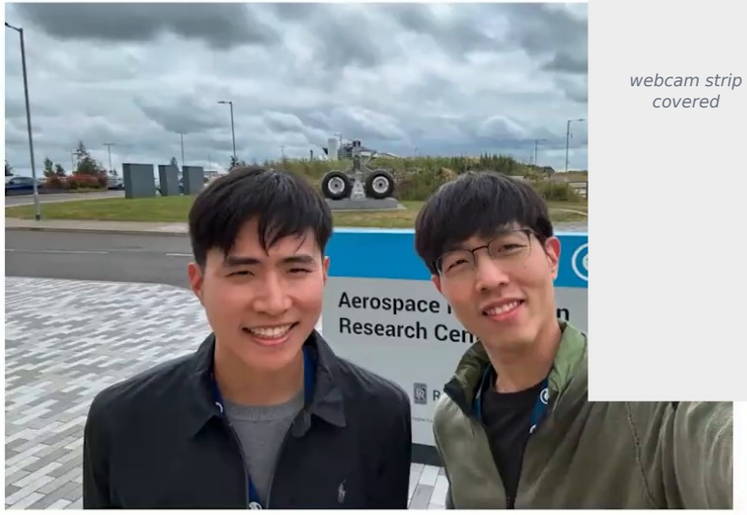
ACKNOWLEDGEMENTS

Other mentors (academia)

▶ 51:08

- Sylvia Herbert (UCSD), Donggun Lee (NCSU), Somil Bansal (Stanford), Namhoon Cho (Cranfield).
- Sylvia's openness during his undergrad project led him to choose Berkeley.





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ACKNOWLEDGEMENTS

Mentors in the later PhD

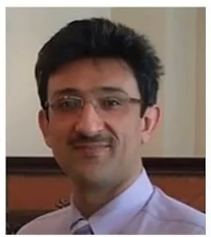
▶ 51:41

- Photos with Somil and Namhoon, who mentored him in the later years of the PhD.

Other Mentors (Industry)



Shaun Mcwherter
(NASA)



Mir Jalali
(Joby Aviation)

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ACKNOWLEDGEMENTS

Other mentors (industry)

▶ 51:58

- Shaun McWhorter (NASA) and Mir Jalali (Joby Aviation). He learned the real-world industry challenges, including during a Joby internship.

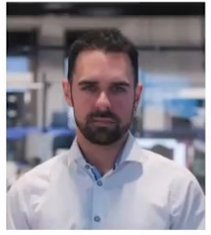
Acknowledgements



Negar Mehr
& ICON Lab



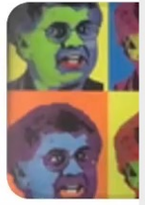
Andrea Bajscy



Alonso Marco



David
Fridovich-Keil



Sastry
Group

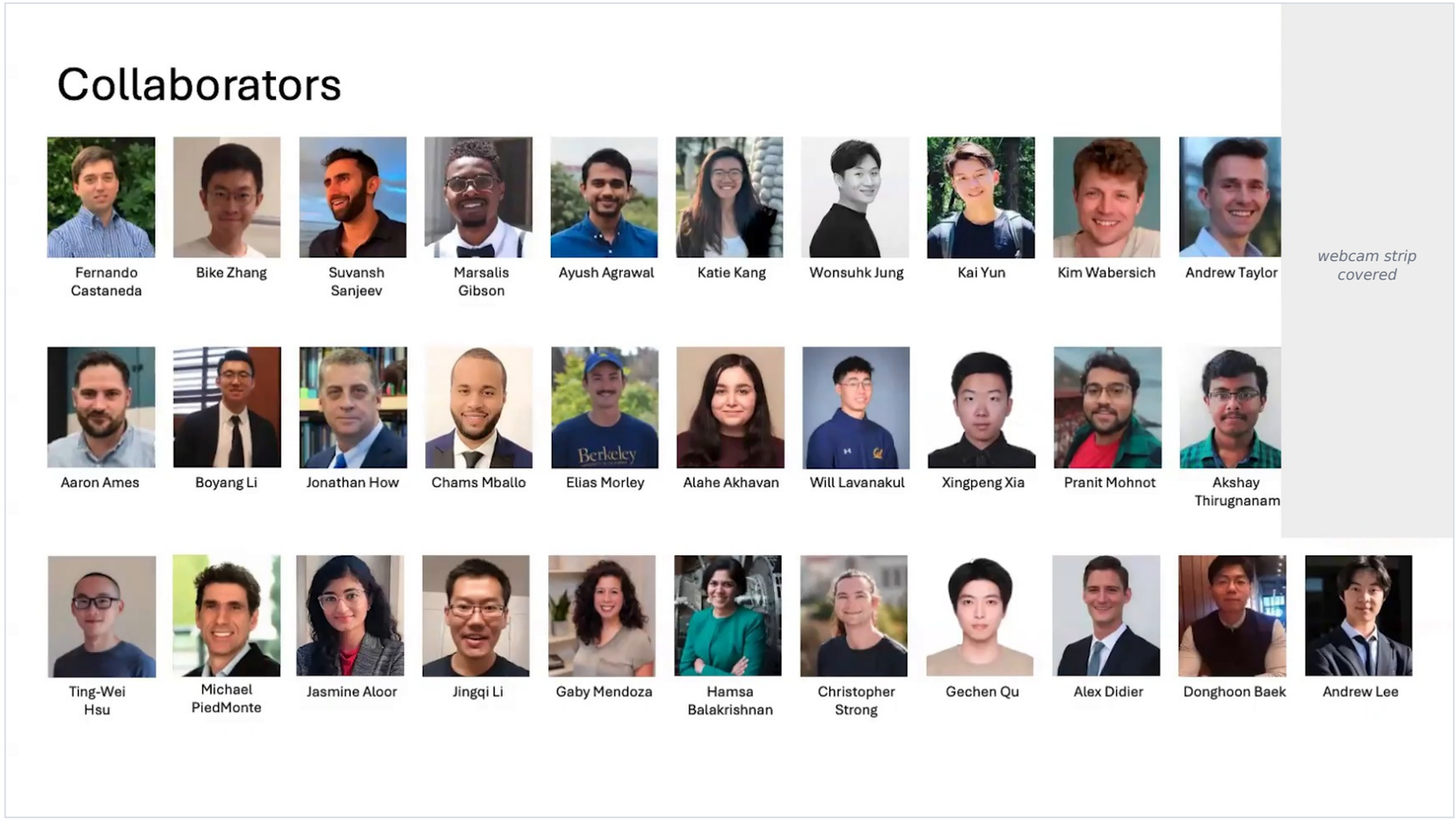
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ACKNOWLEDGEMENTS

Acknowledgements

▶ 52:13

- Informal mentors: Negar Mehr and the ICON Lab, Andrea Bajcsy, Alonso Marco, David Fridovich-Keil, and the Sastry group.

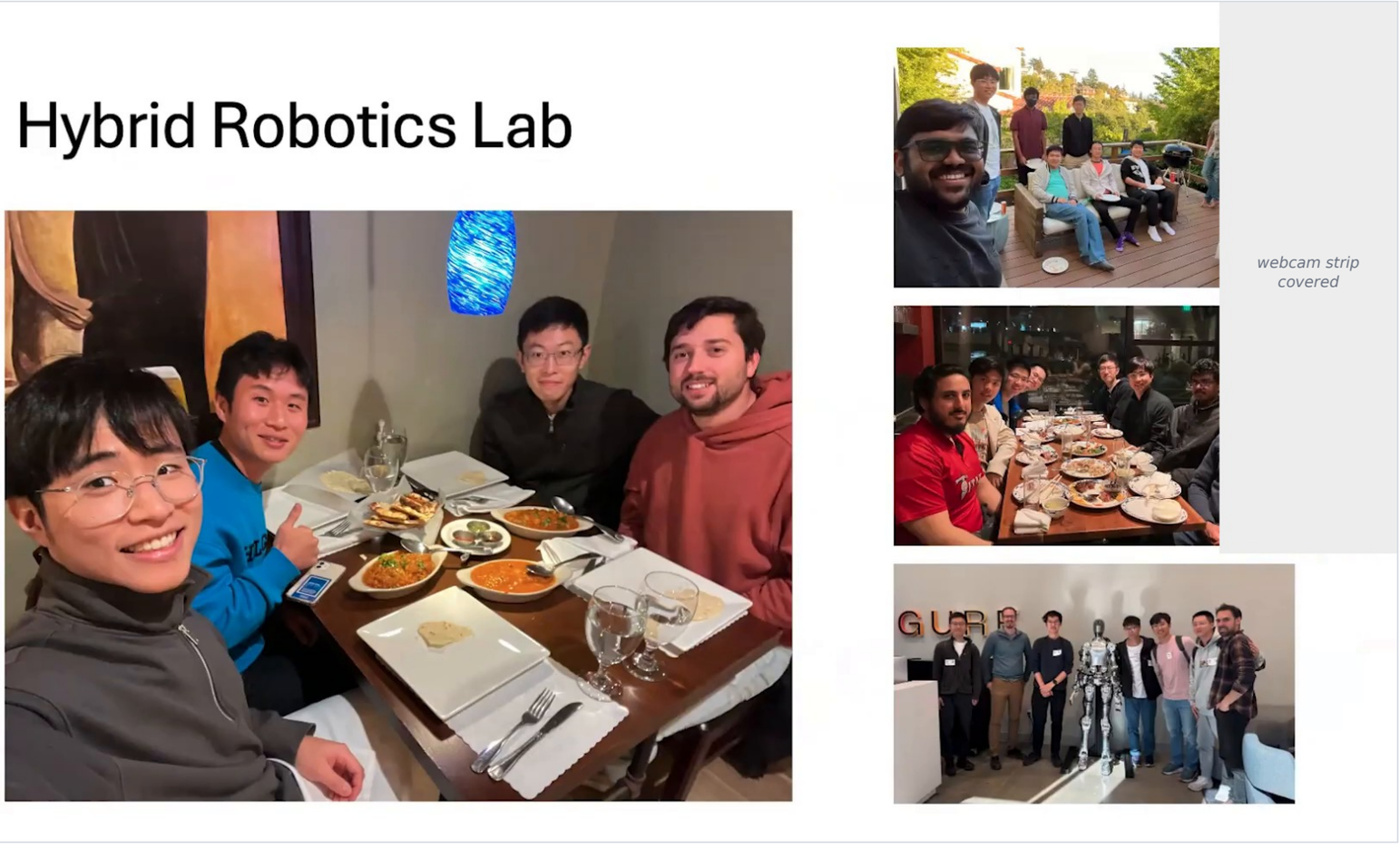


ACKNOWLEDGEMENTS

Collaborators

▶ 52:39

- Thanks to all co-authors and collaborators. None of the work shown was possible without them.



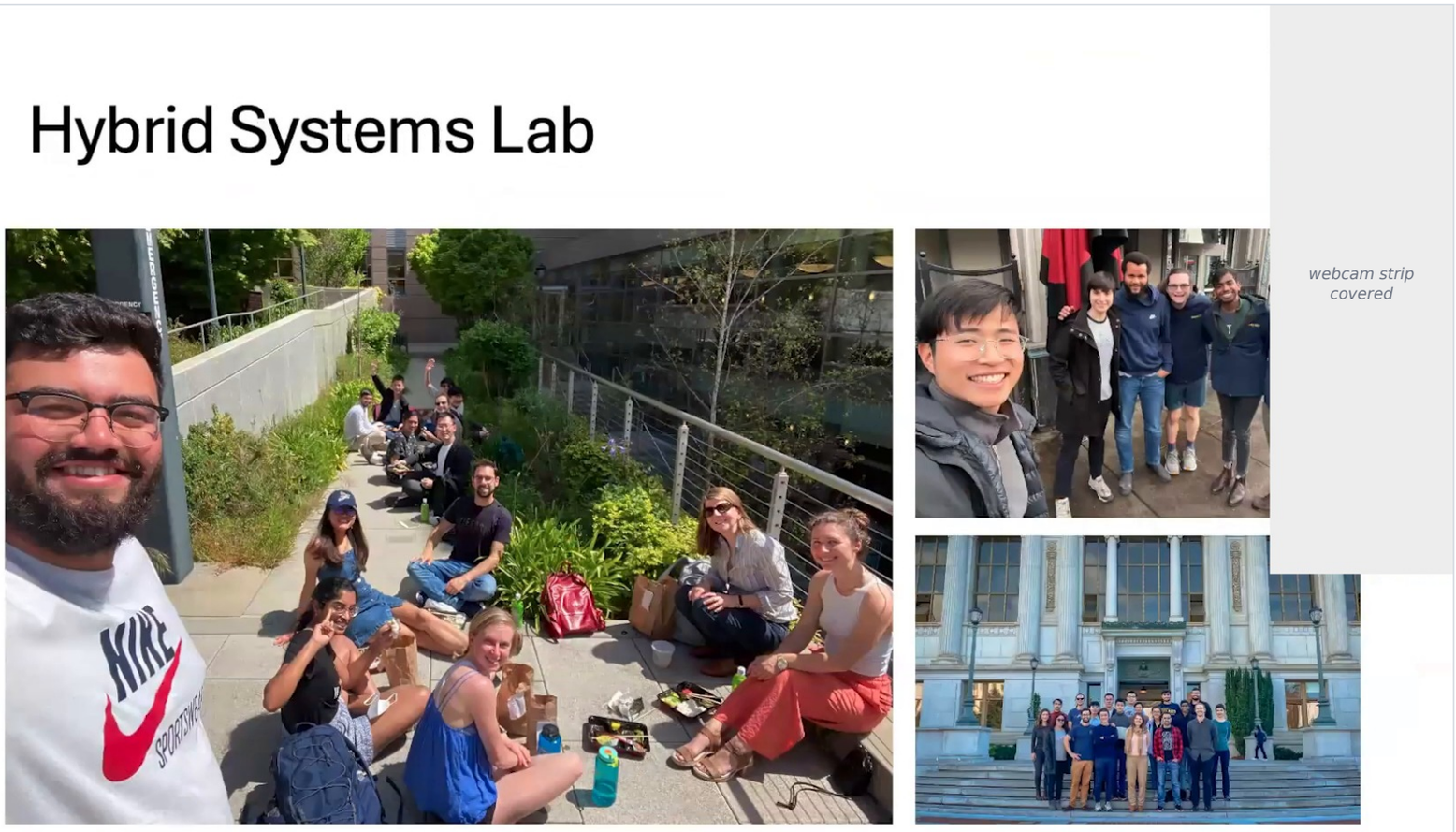
webcam strip covered

ACKNOWLEDGEMENTS

Hybrid Robotics Lab

▶ 52:50

- Friends in Koushil Sreenath's lab.



ACKNOWLEDGEMENTS

Hybrid Systems Lab

▶ 52:56

- Friends in Claire Tomlin's lab.



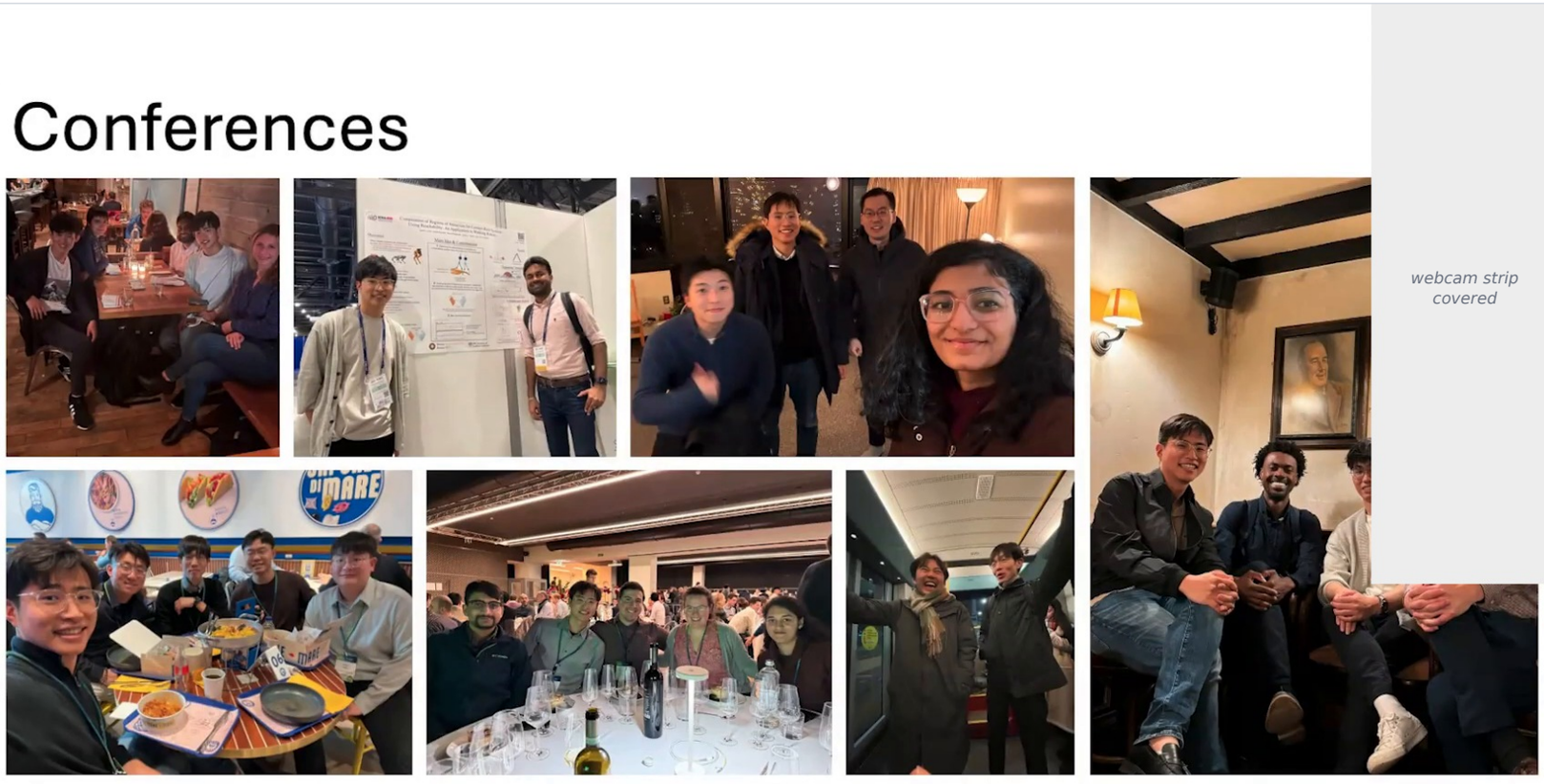
webcam strip covered

ACKNOWLEDGEMENTS

Both labs together

▶ 52:59

- The two labs together at a celebration hosted by his advisors.

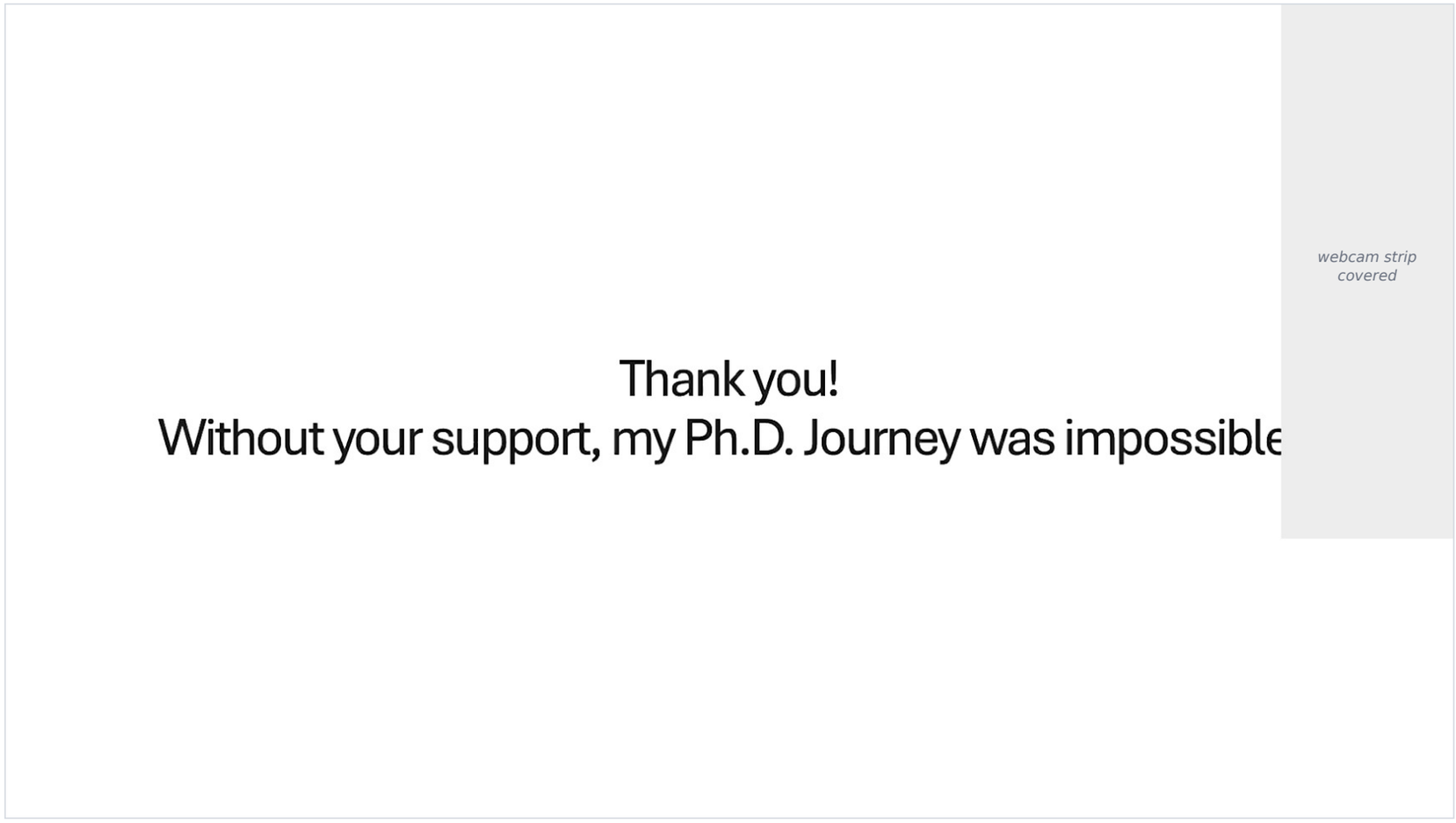


ACKNOWLEDGEMENTS

Conferences

▶ 53:09

- Time with friends from other institutions at conferences.
- Next comes a note that acknowledgements to friends and family were left out of the video for privacy.



ACKNOWLEDGEMENTS

Thank you

▶ 53:35

- Closing slide: "Thank you! Without your support, my Ph.D. Journey was impossible."